Networked and Distributed Control

Stability under Delay and Consensus Based Distributed Optimization

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Control under Delay

With a = 5, b = 9 and c = 1 the system is defined as

$$A = \begin{pmatrix} 0 & -0.5 \\ -3.8 & -1 \end{pmatrix} \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{1.1}$$

1.1. Question 1 - Base Case State-Feedback Control $\bar{K}$

First the simple case of placing poles of a full-state information controller is tasked. The closed loop of the system is described by the dynamics Equation 1.3. The poles are placed such that $A - B\hat{K}$ has the eigenvalues of $-2 \pm j$ to gain $\bar{K} = \begin{pmatrix} 3 & -1.026 \end{pmatrix}$ The initial sampling time was chosen to be $T_s = \pi/(8\omega_n)$ and the range of sampling periods was tested Figure 1.1. The figure shows the the ideal sampling time is between 0.43 and 0.45. The system becomes unstable if the sampling time is higher than $0.6725s$. The effect of increasing spectral radius for decreasing sampling time is in line with the expectation as faster sampling should converge to the continuous system behaviour and since the sampling happens at a higher frequency the eigenvalues need to be closer of the unit circle to result in the same dynamics.

$$\dot{x} = Ax + Bu \tag{1.2}$$

with ZOH sampling and applying $u = \bar{K}x_k$ one step evolution is described by:

$$x_{k+1} = e^{Ah}x_k - \int_0^h e^{At}B\bar{K}x_k dt = \left(e^{Ah} - \left(e^{Ah} - I\right)A^{-1}B\bar{K}\right)x_k = F(h)x_k \tag{1.3}$$

If the input u_k is desired to be separable in the dynamics it can be written as

$$x_{k+1} = F_1(h)x_k + G_1(h)u_k \tag{1.4}$$

with

$$F_1(h) = e^{Ah} \tag{1.5}$$

and

$$G_1(h) = \left(e^{Ah} - I\right)A^{-1}B \tag{1.6}$$

1.2. Question 2 - Small Delay

1.2.1. Analysing stability of a system with a small, constant delay

The effect of adding a small, constant delay is analysed. A small delay is smaller than the sampling time so only the most recent control input u_{k-1} and u_k is present in the dynamics. The dynamics of the system can be rewritten as: the evolution of the autonomous system remains unchanged, the result of application of the previous control input u_{k-1} up until the delay, and the current control input u_k from the delay up until the end of the time interval h. Since the system is linear we can separately compute these elements and add them together.

$$x_{k+1} = e^{AT_s}x_k + \left(e^{AT_s} - e^{A(T_s-\tau)}\right)A^{-1}Bu_k + \left(e^{A(T_s-\tau)} - I\right)A^{-1}Bu_{k-1} \tag{1.7}$$

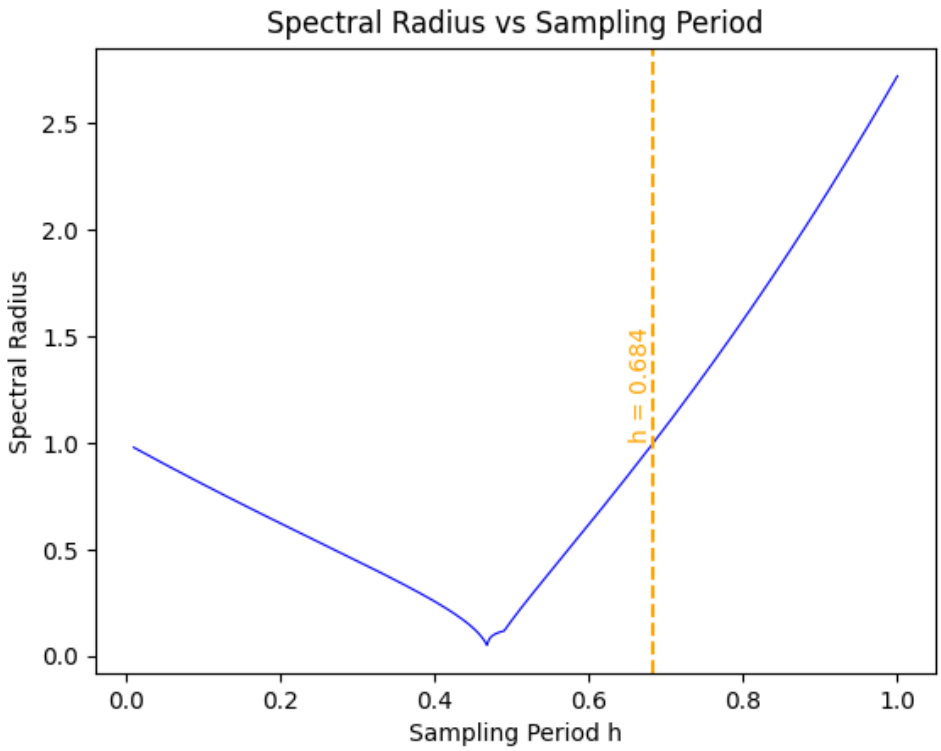

Figure 1.1: Stability of continuos controller vs sampling time

Static feedback controller is applied with state feedback gain of $\bar{K}$, meaning that $u_k = \bar{K}x_k$ and $u_{k-1} = \bar{K}x_{k-1}$. Meaning that the evolution of one time step can be expressed as:

$$x_{k+1} = e^{AT_s}x_k + \left(e^{AT_s} - e^{A(T_s-\tau)}\right)A^{-1}B\bar{K}x_k + \left(e^{A(T_s-\tau)} - I\right)A^{-1}\bar{K}x_{k-1} \tag{1.8}$$

Stacking vertically the states x_k and x_{k-1} to get x_k^e allows us to rewrite the system in the standard form of

$$\begin{pmatrix} x_{k+1} \\ x_k \end{pmatrix} = \begin{pmatrix} e^{Ah} - \left(e^{Ah} - e^{A\tau}\right)A^{-1}B\bar{K} & -\left(e^{A\tau} - I\right)A^{-1}B\bar{K} \\ I & 0 \end{pmatrix}\begin{pmatrix} x_k \\ x_{k-1} \end{pmatrix} \tag{1.9}$$

which compactly expressed as

$$x_{k+1}^e = F^e(h,\tau)x_k^e \tag{1.10}$$

This system is equivalent to the separated form of input and states as:

$$x_{k+1}^e = \begin{pmatrix} e^{Ah} & 0 \\ I & 0 \end{pmatrix}x_k^e + \begin{pmatrix} -\left(e^{Ah} - e^{A\tau}\right)A^{-1}B & -\left(e^{A\tau} - I\right)A^{-1}B \\ 0 & 0 \end{pmatrix}u_k^e \tag{1.11}$$

with

$$u_k^e = \begin{pmatrix} u_k \\ u_{k-1} \end{pmatrix} \tag{1.12}$$

which can be compactly expressed as

$$x_{k+1}^e = F_2(h)x_k^e + G_2(h,\tau)u_k^e \tag{1.13}$$

In this problem u_k^e can be expressed as

$$u_k^e = -\begin{pmatrix} \bar{K} & 0 \\ 0 & \bar{K} \end{pmatrix}x_k^e = -\bar{K}^e x_k^e \tag{1.14}$$

Stability of the system is analysed via the means of assessing the value of the spectral radius of $F^e(h,\tau)$. This is possible due to having constant sampling time h and delay τ, meaning that the state evolves as:

$$x_k^e = \left[F^e(h,\tau)\right]^k x_0^e \tag{1.15}$$

The spectral radius ρ is equivalent to the absolute value of the largest eigenvalue. It is easy to see that having ρ smaller than 1 results in asymptotic stability. It should also be kept in mind that this derivation is only valid to delays that are smaller than the sampling time, therefore only these possibilities are examined. The spectral radius with respect to h and τ is indicated if it is smaller than on Figure 1.2

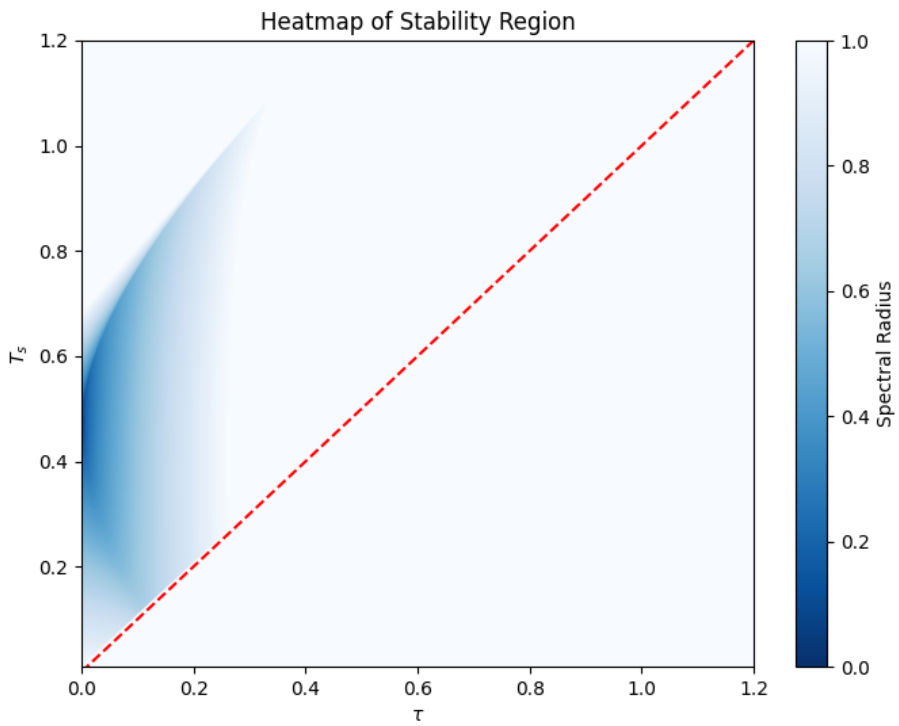

Figure 1.2: T_s and τ vs spectral radius

The plot corresponds to the previously obtained result of stability under sampling but it also produces additional insight. It shows that under a small delay the stable maximum sampling increases and that the delay the system can handle is not strongly dependent on the sampling time with a maximum value of around $\tau = 0.248$. The red line indicates that only the region to the left of it is modelled.

1.2.2. Optimizing K_u for Robustness for Delay

Next the task is to design a dynamic controller that performs better under delays with a fixed sampling time of $h = 0.5$. In this problem a dynamics controller is considered with introducing a new term that adds the to the current input based on the previous input. The previously deployed $\bar{K}$ is continued to being used. This is expressed as:

$$u_k = -\bar{K}x_k + K_u u_{k-1} \tag{1.16}$$

This results in the stacked form of:

$$u_k^e = -\begin{pmatrix} \bar{K} & 0 \\ 0 & \bar{K} \end{pmatrix} x_k^e + K_u u_{k-1}^e \tag{1.17}$$

This is incorporated in the state evolution equation as

$$\begin{pmatrix} x_{k+1}^e \\ u_k^e \end{pmatrix} = \begin{pmatrix} F_2(h) & G_2(h,\tau) \\ 0 & I \end{pmatrix} \begin{pmatrix} x_k^e \\ u_k^e \end{pmatrix} \tag{1.18}$$

$$= \begin{pmatrix} F_2(h) & G_2(h,\tau) \\ 0 & I \end{pmatrix} \begin{pmatrix} x_k^e \\ -k\bar{K}^e x_k^e + K_u u_{k-1}^e \end{pmatrix} \tag{1.19}$$

$$= \begin{pmatrix} F_2(h) & G_2(h,\tau) \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -\bar{K}^e & K_u \end{pmatrix} \begin{pmatrix} x_k^e \\ u_{k-1}^e \end{pmatrix} \tag{1.20}$$

$$= \begin{pmatrix} F_2(h) - G_2(h,\tau)\bar{K}^e & G_2(h,\tau)K_u \\ -\bar{K}^e & K_u \end{pmatrix} \begin{pmatrix} x_k^e \\ u_{k-1}^e \end{pmatrix} \tag{1.21}$$

Which is equivalent to the form where u_{k-1} is stored instead of expressed as the function of the previous states

$$\begin{pmatrix} x_{k+1} \\ u_k \end{pmatrix} = \begin{pmatrix} e^{Ah} & \left(e^{Ah} - e^{A\tau}\right) A^{-1}B & \left(e^{A\tau} - I\right) A^{-1}B \\ 0 & I & 0 \end{pmatrix} \begin{pmatrix} x_k \\ u_k \\ u_{k-1} \end{pmatrix} \tag{1.22}$$

$$= \begin{pmatrix} e^{Ah} & \left(e^{Ah} - e^{A\tau}\right) A^{-1}B & \left(e^{A\tau} - I\right) A^{-1}B \\ 0 & I & 0 \end{pmatrix} \begin{pmatrix} x_k \\ -\bar{K}x_k + K_u u_{k-1} \\ u_{k-1} \end{pmatrix} \tag{1.23}$$

$$= \begin{pmatrix} e^{Ah} & \left(e^{Ah} - e^{A\tau}\right) A^{-1}B & \left(e^{A\tau} - I\right) A^{-1}B \\ 0 & I & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ -\bar{K} & K_u \\ 0 & I \end{pmatrix} \begin{pmatrix} x_k \\ u_{k-1} \end{pmatrix} \tag{1.24}$$

$$= \begin{pmatrix} e^{Ah} - \left(e^{Ah} - e^{A\tau}\right) A^{-1}B\bar{K} & \left(e^{Ah} - e^{A\tau}\right) A^{-1}BK_u + \left(e^{A\tau} - I\right) A^{-1}B \\ -\bar{K} & K_u \end{pmatrix} \begin{pmatrix} x_k \\ u_{k-1} \end{pmatrix} \tag{1.25}$$

$$= F^u(h,\tau) \begin{pmatrix} x_k \\ u_{k-1} \end{pmatrix} \tag{1.26}$$

Following the same argument as previously, the value of $F^u(h,\tau)$ is constant due to constant h, and τ, therefore the stability of the system can be assessed via the spectral radius. The value of $F^u(h,\tau)$ also depends on K_u, therefore the stability of the system can be mapped to different values of K_u.

Following the intuition to derive a stabilizing K_u term didn't result in any meaningful solution, as the matrices of the dynamics non-linearly depend on the τ. Thus an optimization of the system was done to find suitable K_u values. Examining a range of K_u value between 0 and 1 lead to the realisation that increasing K_u does allow for a higher delay tolerance but it doesn't preserve stability for the system without delay. This phenomena is shown for $K_u \in [-1, 1]$ on Figure 1.3. It shows that having negative K_u helps with increasing the robustness with respect to τ. Based on the plot the optimal value for K_u was chosen such that it preserves the stability for the zero-delay case while maximizes which occurs at $K_u = -0.25$. Having a negative value for K_u is cancelling the affect of the previous control action being applied, therefore it makes sense that the higher the absolute value of it while negative the more it benefits robustness against delay while if a point is chosen such that it increases the maximum acceptable delay it doesn't preserve performance at lower, or zero-, delay.

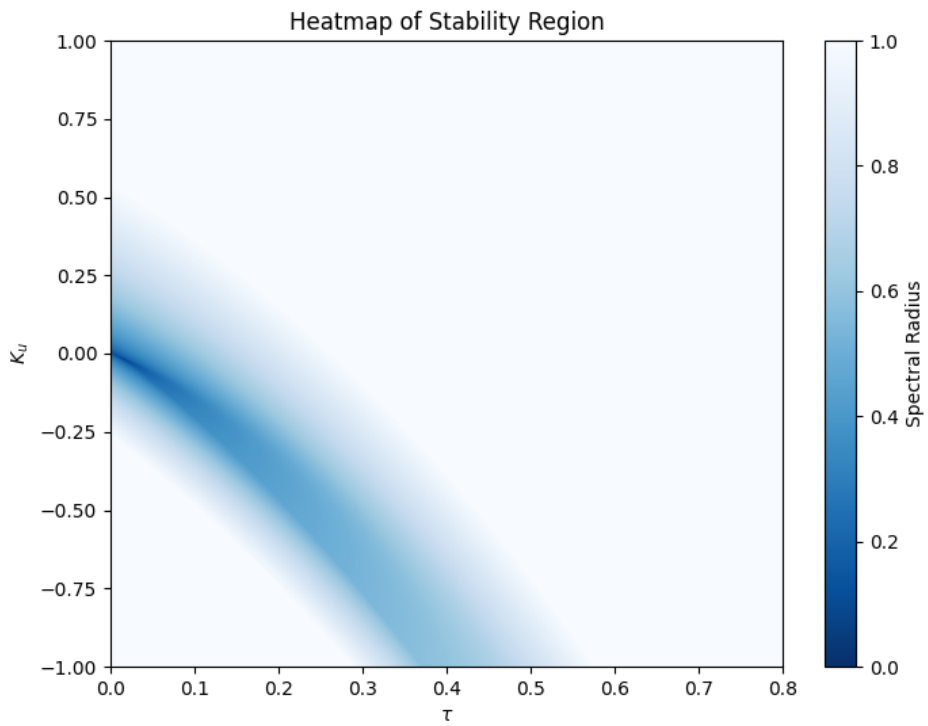

Figure 1.3: K_u vs stable τ region with $h = 0.5$

1.3. Question 3 - Memory Controller

In this section the dynamics of the system remains the same but a dynamic controller is applied, meaning that it uses not only the most recent measurements, but the one taken one time step earlier as well. The measurements are taken at $t = s_k$ but they are only applied in the control at $t = s_k + \tau$. Using ZOH the control action remains constant on the interval $[s_k + \tau, s_{k+1} + \tau)$. In addition to that the dynamic controller is applied $u_k = -\bar{K}x_k - \bar{K}_2 x_{k-1}$ where $\bar{K}_2$ is computed by the place command to place the poles of the original system at -1 and -3 with $\bar{K}_2 = \begin{pmatrix} 3 & -0.5 \end{pmatrix}$ The evolution of the state in the interval $[s_k\ ,\ s_{k+1})$ is given by Equation 1.28.

$$x(s_{k+1}) = e^{Ah}x(s_k) + (e^{A\tau} - I)A^{-1}Bu_{k-1} + (e^{Ah} - e^{A\tau})A^{-1}Bu_k \tag{1.27}$$

This can be rewritten in the form

$$x(s_{k+1}) = e^{Ah}x(s_k) + \begin{pmatrix} (e^{Ah} - e^{A\tau})A^{-1}B & (e^{A\tau} - I)A^{-1}B \end{pmatrix} \begin{pmatrix} u_k \\ u_{k-1} \end{pmatrix} \tag{1.28}$$

with defining matrixes: $G_{11} := (e^{Ah} - e^{A\tau})A^{-1}B$ and $G_{12} := (e^{A\tau} - I)A^{-1}B$ Expanding on the control input as it is defined by the pole placement the system gained is represented as

$$\begin{pmatrix} x(s_{k+1}) \\ x(s_k) \\ x(s_{k-1}) \end{pmatrix} = \begin{pmatrix} e^{Ah} - G_{11}\bar{K} & -G_{11}\bar{K}_2 - G_{12}\bar{K} & -G_{12}\bar{K}_2 \\ I & 0 & 0 \\ 0 & I & 0 \end{pmatrix} \begin{pmatrix} x(s_k) \\ x(s_{k-1}) \\ x(s_{k-2}) \end{pmatrix} \tag{1.29}$$

1.3.1. Stability analysis

This control strategy results in a very poor performing system with stability only occurring at very high sampling frequency and very low disturbance, but with a spectral radius close to 1. It makes the system perform worse than with no design for delay. The stability is shown on Figure 1.5. It shows that the performance of the previous K_u design is much better

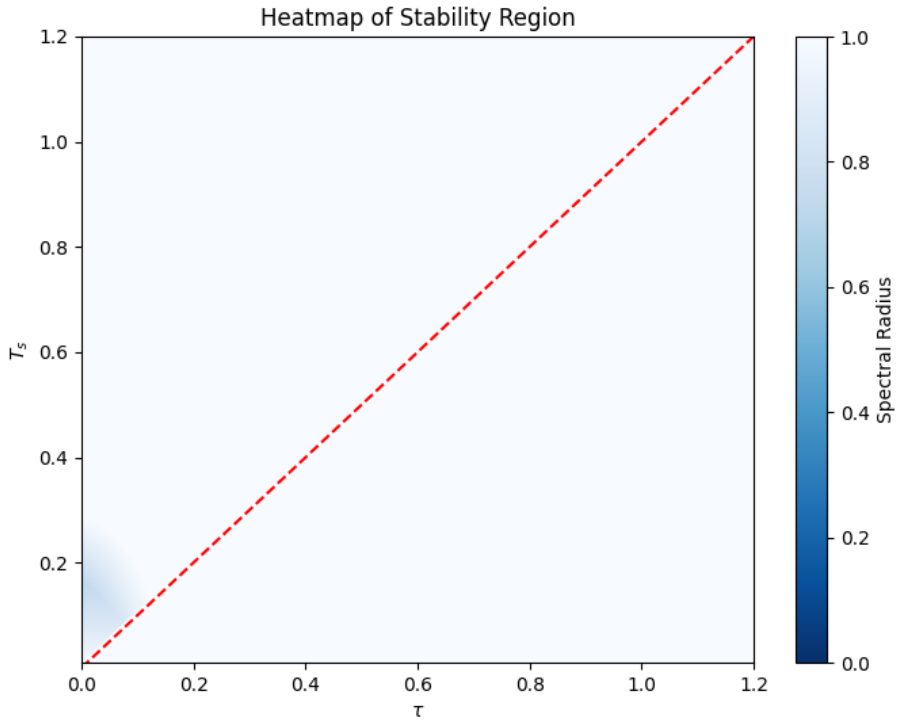

Figure 1.4: Stability Region with Controller $\tilde{K}$ and $\tilde{K}_2$ defined by placed poles

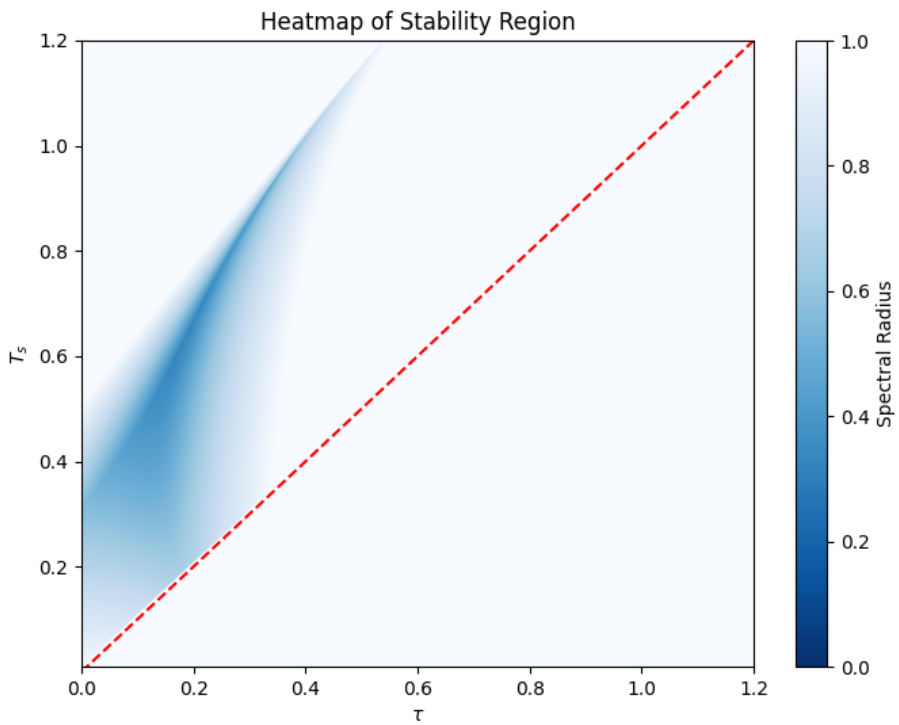

Figure 1.5: Stability Region with Controller $\tilde{K}$ and K_u from optimizing K_u and placing the poles of $\tilde{K}$

1.3.2. Redesigning $\tilde{K} = \begin{pmatrix} \tilde{K}_1 & \tilde{K}_2 \end{pmatrix}$

In this subsection the full-state information feedback is redesigned to optimize for performance under delay. The new weight matrices are: $\tilde{K}_1$ and $\tilde{K}_2$. Since the matrices are changing for different delays,

I didn't see a way to derive optimality. The values of the feedback gains are searched via numerical optimization for a set $h = 0.5$ which produces the optimal gains of Equation 1.30. The newly obtained $\tilde{K}$ gain result in a performance shown on Figure 1.6. It shows that the new design is stable under higher delays while maintaining the performance of the previous design with respect to sampling time. Both designs have a specific h, τ combination that stretches out, therefore for different applications it is not clear which one performs better. In addition to that the previous design has a stronger stability in it's stability region, meaning a lower spectral radius. If the application also includes disturbances that would provide another reason to choose the K_u design over the current one. The reason behind is suspected to be that using one measurement taken one step previously introduces information that is outdated which doesn't mitigate the information loss by experiencing the delay.

$$\tilde{K}_1 = \begin{pmatrix} 1.4004 & -2.0034 \end{pmatrix} \tag{1.30}$$

$$\tilde{K}_2 = \begin{pmatrix} -1.6272 & 1.0287 \end{pmatrix} \tag{1.31}$$

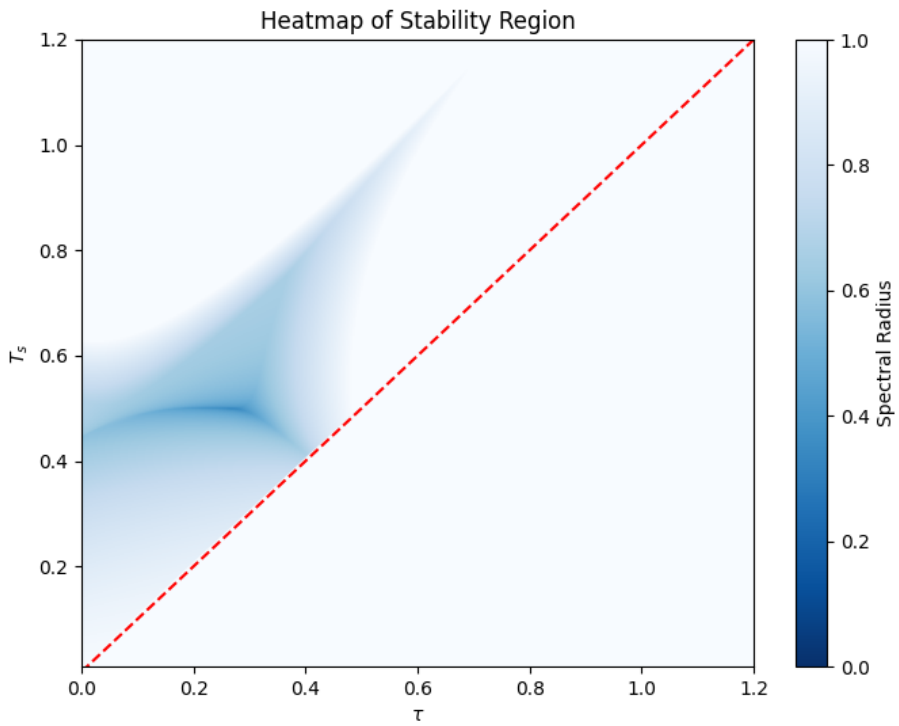

Figure 1.6: Stability Region with Optimized Controller $\hat{K}$ and $\hat{K}_2$

1.4. Question 4 - Time-Variant Sampling

In this section we consider no delays present in the system but the sampling period h is not constant, therefore the k-th time step is denoted by $h_k \in \mathcal{H}$ with $\mathcal{H} = \{h_1, h_2\}$. Two control actions are respectively associated with the sampling times: $\bar{K}$ and $\bar{K}_2$, where the feedback gains have been determined previously by pole placement. Both control actions are static.

The sampling time is picked at random therefore the system becomes a switching system. This results in the loss of linearity, therefore spectral analysis does not correspond to the stability of the system. The system needs to be analysed as a non-linear one to draw conclusions on its stability. To prove stability Lyapunov functions can be used. If the Lyapunov function is contains the evolution of the system, then the convergence properties of the Lyapunov function are also true for the system, proving stability of the origin.

The origin can be stable or attractive. The difference being that for stable systems any arbitrary small $\in$ can be achieved in finite time steps, while for an attractive origin the limit of the states taken k to infinity is 0. The commonality of the properties is that there exists a converging scalar function that bounds the distance of the states from the origin. Furthermore the system can be uniformly stable, which means that the bounding function is independent of the initial values of the states.

Using a Lyapunov function bounds the evolution of the system with a Lyapunov function being strictly increasing with respect to the states but strictly decreasing with respect to the time evolution of the function via the state evolution, meaning the application of chain rule and the time difference function defining the system $x_{k+1} = f(x_k, k)$.

Using positive definite matrices we can easily represent quadratic functions that are strictly increasing or decreasing, therefore based on slide 50 of lecture 2, Global Asymptotic Stability (GAS) of a system can be proven by finding a positive definite solution $P \succ 0$ and $Q \succ 0$ with both $\in S^n$, with S^n being the set of $n \times n$ symmetric matrices, to the Linear Matrix Inequality (LMI) for all $h \in \mathcal{H}$:

$$\left(F(h) - G(h)\bar{K}\right)^T P \left(F(h) - G(h)\bar{K}\right) - P \preceq -Q \tag{1.32}$$

Since in the current case the control action is linked to the sampling time the LMIs to be solved are:

$$\left(F(h_1) - G(h_1)\bar{K}\right)^T P \left(F(h_1) - G(h_1)\bar{K}\right) - P \preceq -Q \tag{1.33}$$

$$\left(F(h_2) - G(h_2)\bar{K}_2\right)^T P \left(F(h_2) - G(h_2)\bar{K}_2\right) - P \preceq -Q \tag{1.34}$$

In order to find a non-zer solution the problem is reformulated as:

$$\left(F(h_1) - G(h_1)\bar{K}\right)^T P \left(F(h_1) - G(h_1)\bar{K}\right) - P \prec -Q - I \tag{1.35}$$

$$\left(F(h_2) - G(h_2)\bar{K}_2\right)^T P \left(F(h_2) - G(h_2)\bar{K}_2\right) - P \prec -Q - I \tag{1.36}$$

The matrices are found which satisfies the stability of the system. The matrices found are:
Matrix P:

$$P = \begin{pmatrix} 1.923357 & -0.2173049 \\ -0.2173049 & 3.41437417 \end{pmatrix}$$

Matrix Q:

$$Q = \begin{pmatrix} 0.30908772 & -0.02166973 \\ -0.02166973 & 0.21955153 \end{pmatrix}$$

1.4.1. Restricting the switching

In this subsection restricting the switching pattern is concerned with an interest if the stability can be checked without using the LMIs. The intervals are restricted to be either $(h_1)^\omega$ or $(h_1, h_2)^\omega$. The system either evolves with a constant sampling time h_1 or an alternating one with h_1 and h_2. Stability of the first case can be done by checking the spectral radius, as before. In order to check stability of the later case we can express the evolution of the states as:

$$x_{k+2} = \left(e^{A(h_1+h_2)} - (e^{Ah_1} - I)A^{-1}B\bar{K} - (e^{Ah_2} - e^{Ah_2})A^{-1}B\bar{K}_2\right)x_k = \bar{A}_2 x_k \tag{1.37}$$

with

$$x_{k+1} = \left(e^{A(h_1} - (e^{Ah_1} - I)A^{-1}B\bar{K}\right)x_k = \bar{A}_1 x_k \tag{1.38}$$

Given $\bar{A}_1$ or $\bar{A}_2$ have spectral radii smaller than 1 then any combination of multiplying by either $\bar{A}_1$ or $\bar{A}_2$ also has a spectral radius smaller than 1, ensuring stability.

1.4.2. Finding stabilizing h_1 and h_2

To guarantee the stability of the system both systems in Equation 1.37 and Equation 1.38 need to be stable. The second scenario is the simple case of discretization, therefore the answer is the same as question 1: $h_1 \in (0, 0.6725]$. In order to make Equation 1.38 stable, a range the values of h_1 and h_2 are tested and the spectral radius is shown on Figure 1.7

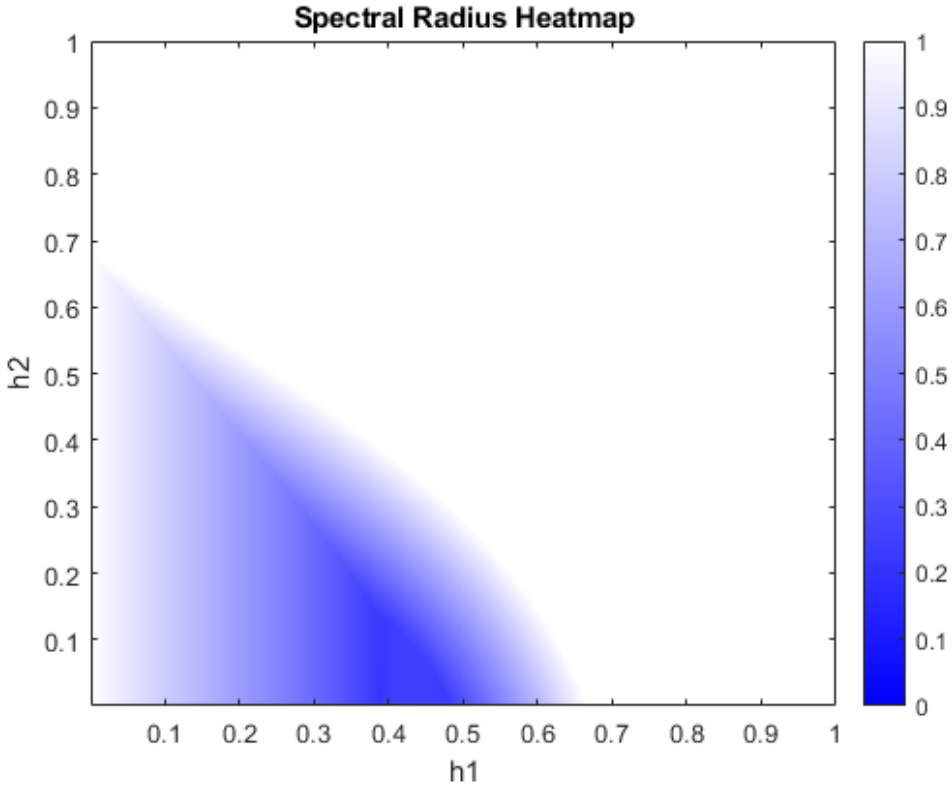

Figure 1.7: Stability Region of h_1 , h_2

Furthermore, simulation of a system can't guarantee it's stability, as different starting values would result in different behaviours and the possibility of having initial state values that the simulation didn't take into account can't be eliminated. However, as the assignment requested it the simulations produces the result of a stable region given 4 initial state vectors in each scenario which are the combinations of values $\frac{1}{2}$ and $-\frac{1}{2}$ for the 2 states shown on Figure 1.8. It shows that the previously obtained stability region is contained in the region given by the simulation. The simulation also provides a much larger stability region, which demonstrates that the method of spectral radius is a conservative method.

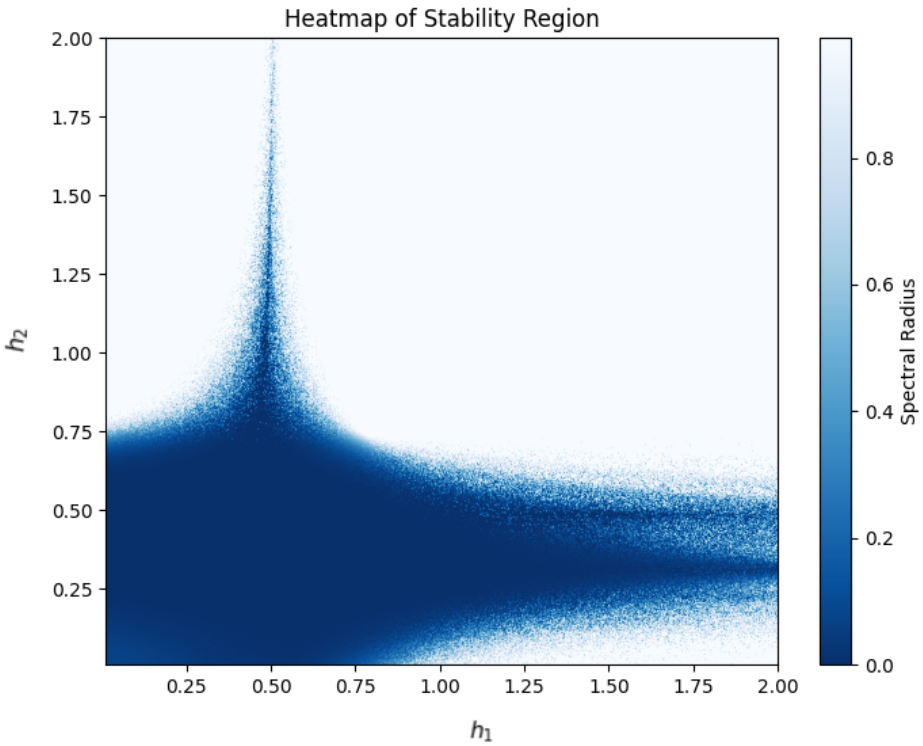

Figure 1.8: Stability region of simulation

2

Control Random Processes under Delay

With a = 5, b = 9 and c = 1 the system is defined as

$$A = \begin{pmatrix} 0 & -0.5 \\ -1 & -2 \end{pmatrix} \tag{2.1}$$

$$B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.2}$$

2.1. Question 1 - Deterministic Switching but Indeterministic Control Strategy

In this section the sampling time is either $h_1 = h$ or $h_2 = 2h$ and the switching of the sampling is restricted to be $(h_1 h_2)^\omega$, but, in contrast to the previous assignment, the control strategy is not fixed to the sampling interval that is chosen. If sampling with h_1 occurs either $\bar{K}$ or $\bar{K}_2$ is applied, in case h_2 occurs, either zero input or $\bar{K}_2$ is used.

2.1.1. Automaton

The first task is to describe the system as a ω Automaton: $M = (Q, \Sigma, \delta, Q_0, F)$. Where the state of the machine describes: the sampling interval, either h_1 or h_2 that is taking place between $t \in (s_k, s_{k+1}]$, and the feedback gain either $\bar{K}$ or $\bar{K}_2$ or 0.

First, all the aforementioned properties of a time step and it's control low are described as the first, second and third element of the state of the automaton and their possible combinations are described as Q:

$$Q = \{(h_1, \bar{K}), (h_1, \bar{K}_2), (h_2, \bar{K}_2), (h_2, 0)\} \tag{2.3}$$

Next the set of discrete inputs is defined as Σ. The input of the automaton is the signal that determined which node the state transition to. As the state describes the current time step, the input determines the dynamics of the subsequent one, defining whether h_1 or h_2 is used, whether $\bar{K}$ or $\bar{K}_2$ is the feedback gain on the most recent state measurement or zero input is used.

$$\Sigma = \{\bar{K}, \bar{K}_2, 0\} \tag{2.4}$$

Next, the transition function is defined, which results on the new state given the input and the previous state.

$$\delta : x_{k+1}^a(x_k^a, u_k) = \begin{cases} (h_1, \bar{K}) & \textit{if } x_k^a = (h_2, any) \ and \ u_k = \bar{K} \\ (h_1, \bar{K}_2) & \textit{if } x_k^a = (h_2, any) \ and \ u_k = \bar{K}_2 \\ (h_2, \bar{K}_2) & \textit{if } x_k^a = (h_1, any) \ and \ u_k = \bar{K}_2 \\ (h_2, 0) & \textit{if } x_k^a = (h_1, any) \ and \ u_k = 0 \end{cases} \tag{2.5}$$

As the sequence starts with h_1, the set of possible starting states are Q_0:

$$Q_0 = \{(h_1, \bar{K}), (h_1, \bar{K}_2)\} \tag{2.6}$$

and as the sequence ends with h_2 the possible final states are F:

$$F = \{(h_2, \bar{K}_2), (h_2, 0)\} \tag{2.7}$$

2.1.2. Dynamics of the sampled system

Each state of the automaton corresponds to one dynamics that takes place. For the state vector of the automaton the corresponding state evolutions describe the dynamics:

$$x(s_{k+1}) = f(x(s_k), x_k^a) = \begin{cases} \left(e^{Ah_1} - (e^{Ah_1} - I)A^{-1}B\bar{K}\right) x(s_k) & \textit{if } x_k^a = (h_1, \bar{K}) \\ \left(e^{Ah_1} - (e^{Ah_1} - I)A^{-1}B\bar{K}_2\right) x(s_k) & \textit{if } x_k^a = (h_1, \bar{K}_2) \\ \left(e^{Ah_2} - (e^{Ah_2} - I)A^{-1}B\bar{K}_2\right) x(s_k) & \textit{if } x_k^a = (h_2, \bar{K}_2) \\ \left(e^{Ah_2}\right) x(s_k) & \textit{if } x_k^a = (h_2, 0) \end{cases} \tag{2.8}$$

2.1.3. Stability of the switching system

In order to check stability of the system LMIs are used. If there is a symmetric positive definite S and Q such that the following LMIs hold for all $A_i \in \{A_1, A_2, A_3, A_4\}$:

$$A_i^\top S A_i - S + Q \prec I \tag{2.9}$$

$$T = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix} = \begin{pmatrix} e^{Ah_1} - (e^{Ah_1} - I)A^{-1}B\bar{K} \\ e^{Ah_1} - (e^{Ah_1} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} - (e^{Ah_2} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} \end{pmatrix} \tag{2.10}$$

Stability is proven at $h = 0.4$ by finding the following S and Q matrices:

$$S = \begin{pmatrix} 0.19863165 & 0.0618567 \\ 0.0618567 & 0.29335848 \end{pmatrix}$$

$$Q = \begin{pmatrix} 0.63227855 & 0.09245861 \\ 0.09245861 & 0.82865347 \end{pmatrix}$$

2.1.4. Stability Range of h

Now different values for h are tested and plotted if the LMI solution exists on Figure 2.1 The plot shows that no symmetric matrices are found as the solution of the LMIs, regardless of the sampling time.

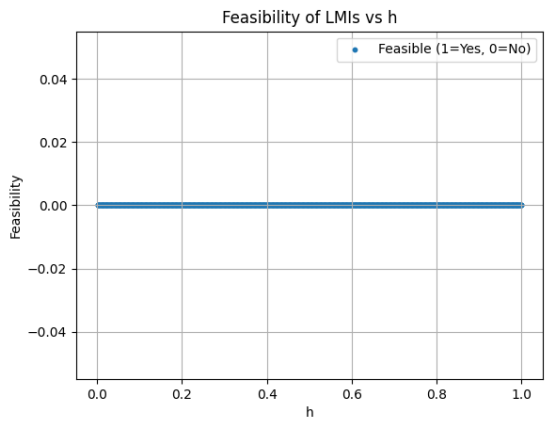

Figure 2.1: Stability Range of h of the Switching system

2.2. Question 2 - Markov Jump Linear Process

2.2.1. Switching Dynamics

The dynamics of the switching is described as a Markov Chain, with probabilities of the state transitions:

- (h_1, any) to $(h_2, \bar{K}_2) = \mathcal{P}(3|1 \ or \ 2) = q$
- (h_1, any) to $(h_2, 0) = \mathcal{P}(4|1 \ or \ 2) = 1 - q$
- (h_2, any) to $(h_1, \bar{K}) = \mathcal{P}(1|3 \ or \ 4) = p$
- (h_2, any) to $(h_1, \bar{K}_2) = \mathcal{P}(2|3 \ or \ 4) = 1 - p$

with $p, q \in (0, 1)$ which then can be expressed as the Markov Process with probability transition P expressing

$$\mathcal{P}(x_{k+1}^a) = \mathcal{P}(x_k^a)P = \mathcal{P}(x_0^a)P^{k+1} \tag{2.11}$$

with P having the value

$$P = \begin{pmatrix} 0 & 0 & q & 1-q \\ 0 & 0 & q & 1-q \\ p & 1-p & 0 & 0 \\ p & 1-p & 0 & 0 \end{pmatrix} \tag{2.12}$$

2.2.2. Expected Evolution Dynamics

The dynamics of the received samples is stochastic, therefore instead of deterministic state values, the expectation of the state is defined. For this the probability of 4 different states at time step k $p_k \in \mathbf{R}^{1\times 4}$, and the 4 outcomes of values of state $x(s_{k+1})$ that correspond to the 4 states of the automaton $E(x_{k+1}^p) \in \mathbf{R}^4$ are defined.

$$E(x(s_{k+1})) = p_{k+1}\, E(x_{k+1}^p) = \sum_{i=1}^{4} p_{k+1}(i)\, E(x_{k+1}^p)(4(i-1) : 4i, 1) \tag{2.13}$$

where p_k evolves as

$$p_{k+1} = p_k P \tag{2.14}$$

and $E(x_{k+1}^p)$ is defined as the inner product between the transition matrix T and expected value of the step

$$E(x_{k+1}^p) = T\, E(x(s_k)) \tag{2.15}$$

where

$$T = \begin{pmatrix} e^{Ah_1} - (e^{Ah_1} - I)A^{-1}B\bar{K} \\ e^{Ah_1} - (e^{Ah_1} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} - (e^{Ah_2} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} \end{pmatrix} \tag{2.16}$$

Putting it all together results in

$$E(x(s_{k+1})) = p_{k+1}\, T\, E(x(s_k)) = \left[\sum_{i=1}^{4} p_{k+1}(i) T(4(i-1) : 4i, :)\right] E(x(s_k)) \tag{2.17}$$

$$E(x(s_{k+1})) = p_k \begin{pmatrix} 0 & 0 & p & 1-p \\ 0 & 0 & p & 1-p \\ q & 1-q & 0 & 0 \\ q & 1-q & 0 & 0 \end{pmatrix} \begin{pmatrix} e^{Ah_1} + (e^{Ah_1} - I)A^{-1}B\bar{K} \\ e^{Ah_1} + (e^{Ah_1} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} + (e^{Ah_2} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} \end{pmatrix} E(x(s_k)) \tag{2.18}$$

which can be rewritten into

$$E(x(s_{k+1})) = p_0 \begin{pmatrix} 0 & 0 & p & 1-p \\ 0 & 0 & p & 1-p \\ q & 1-q & 0 & 0 \\ q & 1-q & 0 & 0 \end{pmatrix}^{k+1} \begin{pmatrix} e^{Ah_1} + (e^{Ah_1} - I)A^{-1}B\bar{K} \\ e^{Ah_1} + (e^{Ah_1} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} + (e^{Ah_2} - I)A^{-1}B\bar{K}_2 \\ e^{Ah_2} \end{pmatrix} E(x(s_k)) \tag{2.19}$$

which means that for k steps taken the evolution of the expected states can be expressed as

$$E(x(s_k)) = p_0 P^k \left(\prod_{i \in K} T p_0 P^{k-i} \right) T E(x(s_0)) \tag{2.20}$$

where $K := \{1, 2..., k-1\}$ is the set of integers from 1 to k-1.

2.2.3. Stability of the Stochastic System

To analyse the stability of the system the following LMIs are used **Markov_Jump**: If there exists a set of positive definite matrices $S_i \succ$ with $i \in \{0, 1, .., N\}$ such that the following gives a positive definite matrix, then the system is stable.

$$S_i - A_i^{\top} \left(\sum_{j=0}^{N} p_{ij} S_j \right) A_i = I \tag{2.21}$$

With $A_i = T_i$, the blocks of the T matrix.

2.2.4. Stability analysis with respect to sampling

The conclusion of question 2 is the stability analysis with respect to sampling time. The Markov Jump process LMIs are used, as for different h values T, and therefore A_i change, therefore the solution to the LMIs is change with respect to the sampling time. The h values ranging from 0.1 to 1 were tested and resulted in Figure 2.2. The analysis shows that the critical sampling time at which the expected stability is not guaranteed is 0.523.

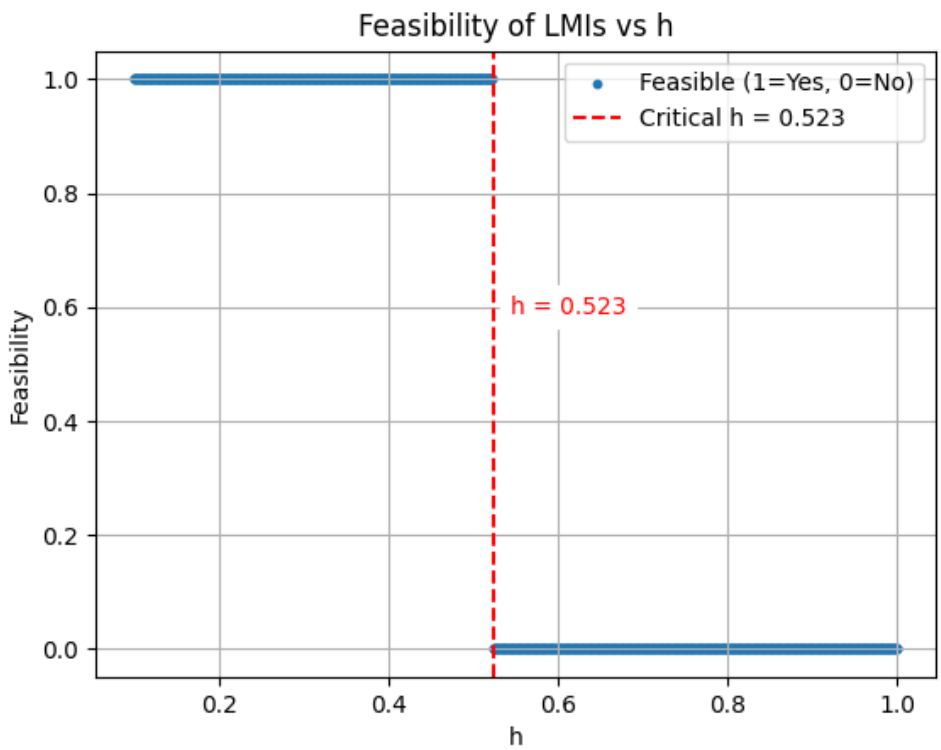

Figure 2.2: Markov Jump Process Stability

2.3. Question-3

In the system with a small delay $\tau \in [0, h)$ and no switching system is analyzed with A and B matrices:

$$A = \begin{pmatrix} -3.7 & 0 \\ 1 & 1.5 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

2.3.1. Stability analysis with respect to sampling and delay

Since in this subproblem the sampling period is uniform the spectral radius can be applied to determine stability. Furthermore, the state-feedback controller $\hat{K}$ is defined by placing the poles at $\{-2+j, -2-j\}$. The evolution of the system under a small delay is expressed as the sum of the autonomous evolution, the evolution that is due the the previous control action acting until τ, and the evolution due to the new current control action acting from time τ until the end of the time step. The controller is not switching, but the sampling time still does. The evolution is then expressed as:

$$x_{k+1} = e^{Ah}x_k - \left(e^{A\tau} - I\right)A^{-1}B\hat{K}x_{k-1} - \left(e^{Ah} - e^{A\tau}\right)A^{-1}B\hat{K}x_k \tag{2.22}$$

with $\hat{K}$ having the value

$$\hat{K} = \begin{pmatrix} 1.8 & 13.25 \end{pmatrix} \tag{2.23}$$

This system is equivalent to the static controlled system with extended states as:

$$\begin{pmatrix} x_{k+1} \\ x_k \end{pmatrix} = \begin{pmatrix} e^{Ah} - \left(e^{Ah} - e^{A\tau}\right)A^{-1}B\hat{K} & -\left(e^{A\tau} - I\right)A^{-1}B\hat{K} \\ I & 0 \end{pmatrix} \begin{pmatrix} x_k \\ x_{k-1} \end{pmatrix} \tag{2.24}$$

which is rewritten into a more concise form of with the extended state x_k^e being the states x_k and x_{k-1} stacked vertically and the new autonomous system dynamics being represented as $\hat{A}$ as:

$$x_{k+1}^e = F^e(h,\tau)x_k^e \tag{2.25}$$

Since the sampling time is switching between $h_1 = h$ and $h_2 = 2h$ two time steps are expressed as repeating steps.

$$x_{k+2}^e = F^e(h,\tau)F^e(2h,\tau)x_k^e \tag{2.26}$$

which then can be rewritten as

$$x_{i+1}^e = F_2^e(h,\tau)x_k^e \tag{2.27}$$

with

$$F_2^e(h,\tau) = F^e(h,\tau)F^e(2h,\tau) \tag{2.28}$$

Under the switching sampling with constant h and constant delay this system has a invariant gain matrix $F_2^e(h,\tau)$, meaning that the spectral radius of $F_2^e(h,\tau)$ provides whether the system explodes the evolution dynamics. If the poles are contained within the unit disk the system is stable. For the range of sampling intervals h and delays τ the spectral radius is tested and if it is lower than 1 they are plotted on Figure 2.3. The analysis shows that the obtained stability region has two distinct segments.

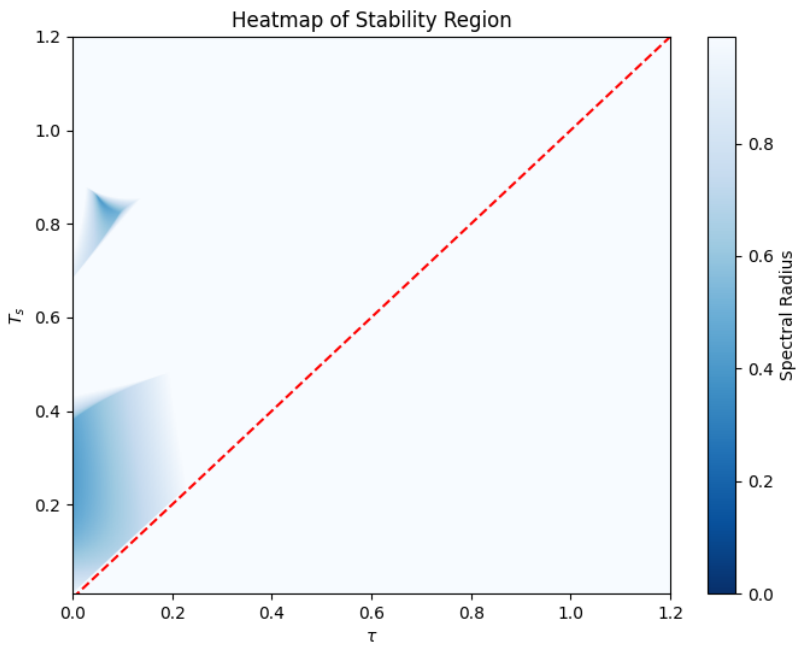

Figure 2.3: Stability with respect to τ and h

2.3.2. Jordan form approach

In this part of the report the following matrices are used:

$$A = \begin{pmatrix} -3.7 & 0 \\ 1 & 1.5 \end{pmatrix} \tag{2.29}$$

$$B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.30}$$

$$\hat{K} = \begin{pmatrix} 1.8 & 13.25 \end{pmatrix} \tag{2.31}$$

This section is based on the slides of Lecture 4 of the course SC42101 at TU Delft. In this subproblem the problem is reformulated as a system with varying delays and sampling intervals such that is contains the previously analysed system. In this method the uncertainness are described as continuos sets, instead of finite ones. To mimic the same problem set up it is described as:

$$h_k \in [h_1, 2h_1] \tag{2.32}$$

$$\tag{2.33}$$

No pocket losses are considered and a small constant delay is considered $\tau \in [0, h_1)$. In this simple case the set of uncertain parameters at time step k is simply $\theta_k = \{h_k\}$. For this system the state evolution is expressed as:

$$x_{k+1} = \Lambda(\theta_k)x_k + M_{-1}u_{k-1} + M_0 u_k \tag{2.34}$$

with

$$\Lambda(\theta_k) = e^{Ah_k} \tag{2.35}$$

$$M_{-1} = \left(e^{A\tau} - I\right)A^{-1}B \tag{2.36}$$

$$M_0 = \left(e^{Ah_k} - e^{A\tau}\right)A^{-1}B \tag{2.37}$$

considering controller $u_k = -\hat{K}x_k$ this can be rewritten with

$$x_k^u = \begin{pmatrix} x_k \\ u_{k-1} \end{pmatrix} \tag{2.38}$$

$$x_{k+1}^u = H(\theta_k)x_k^u \tag{2.39}$$

with

$$H(\theta_k) = \begin{pmatrix} \Lambda(\theta_k) - M_0\hat{K} & -M_1 \\ -\hat{K} & 0 \end{pmatrix} \tag{2.40}$$

This is a linear parameter-varying (LPV) system, where the the uncertain variables describe the system in a non-linear way. The previously deployed LMI method could be used to look for a solution that is proves the existence of a Lyapunov function by solving the LMIs with $S \succ 0$ and $\gamma \in (0,1)$:

$$H^T(\theta_k)SH(\theta_k) - S \preceq -\gamma S \tag{2.41}$$

But since $H(h_k)$ is an element of an infinite set, this problem can not be solved by deploying the inequalities as there are infinitely many of them. The polytopic model takes the infinitely many LMIs and describes a space of matrices that satisfy the infinitely many LMIs. The stability of a polytopic model says (Lecture 4 slides of SC42101):

$$x_{k+1}^e = \left(\sum_{i=1}^{N} \lambda_i H_i\right) x_k^e, \quad \lambda_i \geq 0, \quad \sum_{i=1}^{N} \lambda_i = 1 \tag{2.42}$$

is globally asymptotically stable if there exists $S = S^T \succ 0$, $\gamma \in (0,1)$ such that the following finite set of LMIs are satisfied:

$$H_i^T S H_i - S \preceq -\gamma S, \quad \forall i = 1, \dots, N. \tag{2.43}$$

where H_i are the corner points of the convex polytopic space. By making sure that the matrix S satisfies all the corner points of the polytopic set, all points within the set are also satisfied by the solution S. By making this polytopic smaller we gain a larger solution space for the uncertain values as it gets closer to the underlying LMIs.

The construction of the corner points is the task specific to the problem. The uncertainty space should be a subspace of the convex polytopic space, which has a non-polytopic shape due to the non-linearity of the $H(\theta_k)$, therefore no finite point polytopic set can exactly represent that space. To construct the polytopic, first the parts of $H(\theta_k)$ that are dependent on θ_k are separated as scalar multipliers as:

$$H(\theta_k) = H_0 + \sum_{i=0}^{r} \alpha_i(h_k)F_i \tag{2.44}$$

expanding on $H(\theta_k)$ leads to

$$\begin{pmatrix} e^{Ah_k} - \left(e^{Ah_k} - e^{A\tau}\right)A^{-1}B\hat{K} & -\left(e^{A\tau} - I\right)A^{-1}B \\ -\hat{K} & 0 \end{pmatrix} \tag{2.45}$$

The resulting uncertainty sets are

$$\mathcal{H} = \left\{ H_0 + \sum_{i=0}^{r} \alpha_i(h_k)F_i | h_k \in (h_1, 2h_1) \right\} \tag{2.46}$$

where the terms are defined as

$$H_0 = \begin{pmatrix} e^{A\tau}A^{-1}B\hat{K} & -\left(e^{A\tau} - I\right)A^{-1}B \\ -\hat{K} & 0 \end{pmatrix} \tag{2.47}$$

$$\alpha_0(h_k) = e^{Ah_k} \tag{2.48}$$

$$F_0 = \begin{pmatrix} I - A^{-1}B\hat{K} & 0 \\ 0 & 0 \end{pmatrix} \tag{2.49}$$

$$\tag{2.50}$$

Where e^{Ah_k} should be written in the form where it only depends linearly on e^{h_k}. This is possible via the Jordan form as:

$$e^{Ah_k} = Pe^{h_k}JP^{-1} \tag{2.51}$$

with

$$A = PJP^{-1} \tag{2.52}$$

where

$$J = \begin{pmatrix} -3.7 & 0 \\ 0 & 1.5 \end{pmatrix}$$
$$P = \begin{pmatrix} -5.2 & 0 \\ 1 & 1 \end{pmatrix}$$
$$P^{-1} = \begin{pmatrix} -0.1923 & 0 \\ 0.1923 & 1 \end{pmatrix}$$

With which the system can be redefined in a new coordinate system where $x = Pz$. The stability of the original system is independent to this change therefore the rewriting of the system is not very complicated. It results in the following system.

$$Pz = APz + BKPz \tag{2.53}$$
$$z = P^{-1}APz + P^{-1}Bu_k \tag{2.54}$$

$$\dot{z} = Jz + P^{-1}Bu \tag{2.55}$$

which can be expressed as $\bar{B} = P^{-1}B$ with

$$P^{-1}B = \begin{pmatrix} -0.1923 \\ 0.1923 \end{pmatrix} \tag{2.56}$$

and

$$u_k = \hat{K}x_k = \hat{K}Pz_k = \hat{K}^z z_k \tag{2.57}$$

which leads to the new system:
expanding on $H^z(\theta_k)$ leads to

$$\begin{pmatrix} z_{k+1} \\ u_k \end{pmatrix} = \begin{pmatrix} e^{Jh_k} - \left(e^{Jh_k} - e^{J\tau}\right)J^{-1}\bar{B}\hat{K}^z & -\left(e^{A\tau} - I\right)A^{-1}\bar{B} \\ -\hat{K}^z & 0 \end{pmatrix} \begin{pmatrix} z_k \\ u_{k-1} \end{pmatrix} \tag{2.58}$$

therefore the dynamics can be rewritten with α_0 being a scalar as:

$$H_0^z = \begin{pmatrix} e^{J\tau}J^{-1}\bar{B}\hat{K}^z & -\left(e^{J\tau} - I\right)J^{-1}\bar{B} \\ -\hat{K}^z & 0 \end{pmatrix} \tag{2.59}$$
$$\alpha_0^z(h_k) = e^{Jh_k} \tag{2.60}$$
$$F_0^z = \begin{pmatrix} (I - J^{-1}\bar{B}\hat{K}^z) & 0 \\ 0 & 0 \end{pmatrix} \tag{2.61}$$
$$\tag{2.62}$$

with

$$\alpha_0^z(h_k) = \begin{pmatrix} e^{-3.7h_k} & 0 \\ 0 & e^{1.5h_k} \end{pmatrix} \tag{2.63}$$

which can be represented in R^2 as and produces an example of a result of Figure 2.4

This is a concave function therefore it can be contained by three points as the figure shows, where the third point is the intersection of the tangents Figure 2.5, where the values of h_k are $(0.186302 - 0.2637528)$ respectively for the first and second tangent line.

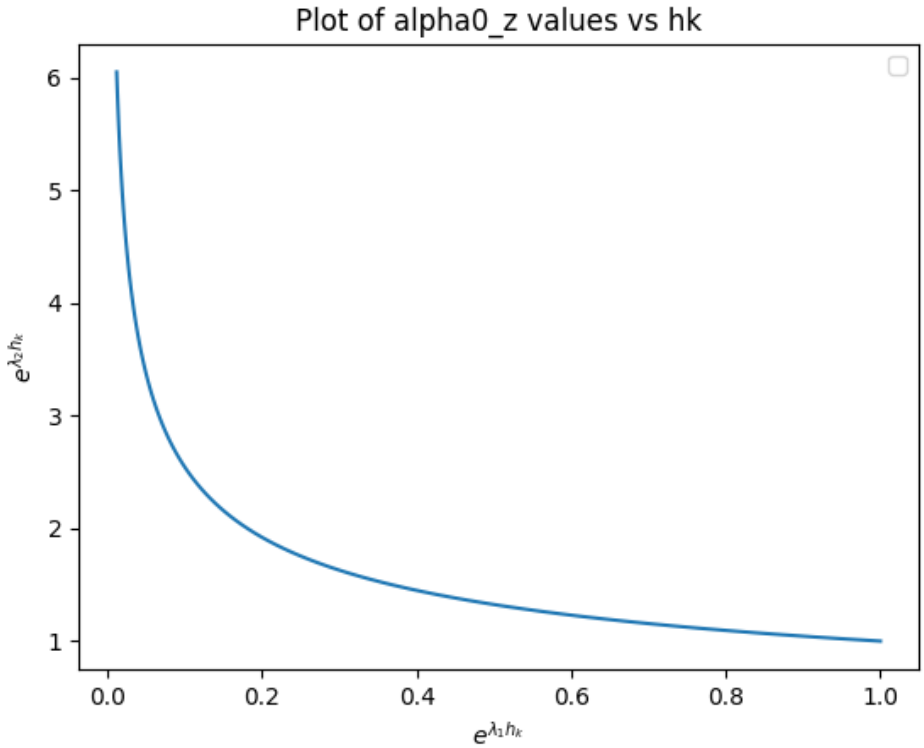

Figure 2.4: α_0^z diagonal elements

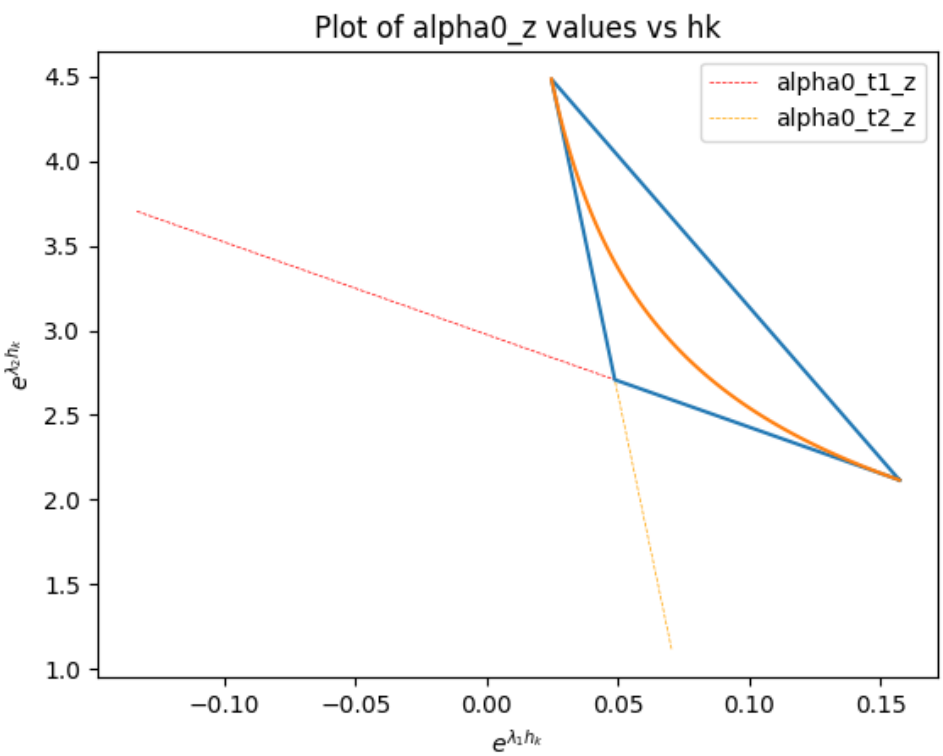

Figure 2.5: Polytope

The three points define the three matrices such as

$$\alpha_{0,1}^{z} = \alpha_{0}^{z}(h1) \tag{2.64}$$
$$\alpha_{0,2}^{z} = \alpha_{0}^{z}(2h1) \tag{2.65}$$
$$\alpha_{0,3}^{z} \tag{2.66}$$

In order to get $\alpha_{0,3}^{z}$ the intersection is found by writing the terms as

$$\alpha_d(h_k) = \begin{pmatrix} e^{-3.7h_k} \\ e^{1.5h_k} \end{pmatrix}, \tag{2.67}$$
$$\alpha_d'(h_k) = \begin{pmatrix} -3.7e^{-3.7h_k} \\ 1.5e^{1.5h_k} \end{pmatrix}. \tag{2.68}$$

Using these, we construct two affine lines:

$$\alpha_d^{t_1}(h_k, h_1) = (h_k - h_1)\alpha_d'(h_1) + \alpha_d(h_1), \tag{2.69}$$
$$\alpha_d^{t_2}(h_k, h_2) = (h_k - h_2)\alpha_d'(h_2) + \alpha_d(h_2). \tag{2.70}$$

These represent tangent line approximations of $\alpha_d(h_k)$ at points h_1 and h_2 respectively. Then the point defined by (h_{k1}, h_{k2}) is defined as

$$\begin{pmatrix} h_{k1} & h_{k2} \end{pmatrix} = (\alpha_d(h2) - \alpha_d(h1)) \begin{pmatrix} \alpha_d'(h_1) \\ -\alpha_d'(h_2) \end{pmatrix}^{-1} \tag{2.71}$$

then the diagonal values of $\alpha_{0,3}^{z}$ can be computed as either $\alpha_d^{t_1}(h_{k1}, h1)$ or $\alpha_d^{t_2}(h_{k2}, h2)$.

2.3.3. Analysing stability with polytope

Following the previously described process if there is a γ value found it is plotted on the heat map as previously for the range of h_1 and τ. Figure 2.6 shows that no stable nodes are found. While the code was running it did find solutions but the iterations were time consuming to the point where these rare occurrences couldn't not be captured in a reasonable time.

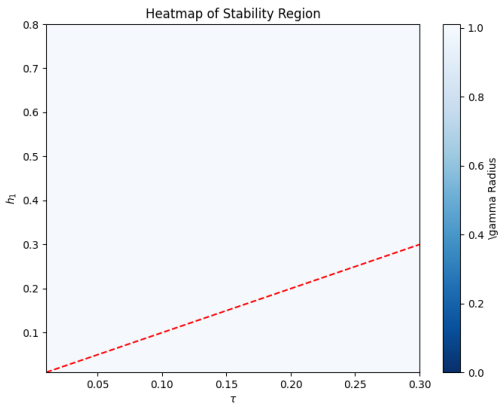

Figure 2.6: Caption

2.3.4. Comparison

The analysis shows that the deployed method is very conservative. using more points to describe the polytope would help to reduce the gap, but that would also increase the computational time it requires significantly. As the set of LMIs to satisfy is larger, the solution exists for a smaller subspace of h_1 and τ.

3

Convergence of Distributed Optimization

3.1. Problem 1 - Optimality Conditions

Given a the definition of the optimum $x^\star$ by Equation 3.6 the problem is find the proof for statement: with $f(x)$ being a convex function and $\mathcal{X}$ being a convex set $x^\star$ is the optimum if and only if Equation 3.2 holds.

$$f(z) - f(x^\star) \geq 0 \; \forall \; z \; \in \; \mathcal{X} \tag{3.1}$$

$$0 \leq (z - x^\star)^\top \nabla f(x^\star) \; \forall \; z \; \in \; \mathcal{X} \tag{3.2}$$

Given the optimum has been reached there are two options, first the point where the function is minimized on the space $\mathcal{R}^n$ is included in the space $\mathcal{X}$, which means that the optimum is given where the gradient is zero, due the the function being convex, meaning that Equation 3.3 holds for all z, therefore both directions of the statement is proven.

$$0 = (z - x^\star)^\top \nabla f(x^\star) \tag{3.3}$$

The other option is that the point where the gradient is zero is not in $\mathcal{X}$. In this case the optimum is on the boundary of $\mathcal{X}$. This case is investigated further in the following sections.

Since the shape of $f(x)$ is give to be quadratic with $H \; \geq \; 0$ Equation 3.4, the gradient is given by Equation 3.5

$$f(x) = \frac{1}{2} x^\top H x + c^\top x \tag{3.4}$$

$$\nabla f(x) = Hx + c \tag{3.5}$$

3.1.1. Necessity

In this section it is shown that given $f(x)$ and $\mathcal{X}$ are convex if $x^\star$ is optimal then Equation 3.2 holds.

z is defined as the convex combination of $x^\star$ and x_b where x_b is on the boundary of $\mathcal{X}$: $x_b \; \in \; \partial\mathcal{X}$: $z = x^\star + \lambda(x_b - x^\star) \mid \lambda \; \in \; [0, 1]$ given $\mathcal{X}$ is convex, $z \; \in \; \mathcal{X}$. Given $x^\star$ is optimal then the following statement holds:

$$f(z) - f(x^\star) \geq 0 \; \forall \; z \; \in \; \mathcal{X} \tag{3.6}$$

Where $f(z)$ can be expressed as:

$$f(z) = f(x^\star) + \int_{x^\star}^{z} \nabla f(x) dx \tag{3.7}$$

Which leads to the optimality expression Equation 3.9:

$$\int_{x^\star}^{z} \nabla f(x) dx \geq 0 \; \forall \; z \; \in \; \mathcal{X} \tag{3.8}$$

Given that $x^\star$ is optimal, this relation always holds. Since all $z \in \mathcal{X}$ can be expressed as a convex combination of x_b and $x^\star$ the expression can be rewritten into

$$\int_{x^\star}^{x^\star+\lambda(x_b-x^\star)} \nabla f(x)dx \geq 0 \; \forall \; z \in \mathcal{X} \tag{3.9}$$

For the infinitely small distance $d\lambda$ the gradient is constant, therefore the integral evaluates as

$$d\lambda(x_b - x^\star)^\top \nabla f(x^\star) \geq 0 \;\; \forall \; x_b \in \partial\mathcal{X} \tag{3.10}$$

taking the integral with respect to any positive $\lambda \in [0,1]$ only scales the expression

$$\int_0^\lambda d\lambda(x_b - x^\star)^\top \nabla f(x^\star) = \lambda(x_b - x^\star)^\top \nabla f(x^\star) \geq 0 \;\; \forall \; x_b \in \partial\mathcal{X}, \; \lambda \in [0,1] \tag{3.11}$$

which means that given $x^\star$ is optimal, the gradient in the direction of $\forall \; z$ has to be positive. If this was not the case there would exist a point in $\mathcal{X}$ in the negative gradient direction where the function would by definition have a lower value by shifting to that direction. Since all possible values of λ are positive they don't change the sign of the relation. Since Equation 3.12 holds by definition of z Equation 3.13 is equivalent to Equation 3.11.

$$z - x^\star = x^\star + \lambda(x_b - x^\star) - x^\star = \lambda(x_b - x^\star) \tag{3.12}$$

$$0 \leq (z - x^\star)^\top \nabla f(x^\star) \; \forall \; z \in \mathcal{X} \tag{3.13}$$

which means that if $x^\star$ is optimal then Equation 3.13 holds.

3.1.2. Sufficiency

In this section it is shown that given $f(x)$ and $\mathcal{X}$ are convex and Equation 3.2 holds then $x^\star$ is optimal.

Due to the first order property of a convex function, $f(z)$ is lower bounded by the tangent plane at any point Networked and Distributed Control Systems Course Team, 2025, this is applied at point $x^\star$.

$$f(z) \geq f(x^\star) + (z - x^\star)^\top \nabla f(x^\star) \tag{3.14}$$

Which can be rewritten in the form

$$f(z) - f(x^\star) \geq (z - x^\star)^\top \nabla f(x^\star) \tag{3.15}$$

given that

$$0 \leq (z - x^\star)^\top \nabla f(x^\star) \tag{3.16}$$

It can be stated that

$$f(z) - f(x^\star) \geq 0 \tag{3.17}$$

Which means that the optimality condition is met.

3.2. Problem 2 - Quadratic Parallel Optimization

Given the function to minimize with symmetric $H \succ 0$

$$V(u_1, u_2) = \frac{1}{2}\begin{pmatrix} u_1^\top & u_2^\top \end{pmatrix}\begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} c_1^\top & c_2^\top \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + d \tag{3.18}$$

Through the process of iterations:

1. initialize at starting point (u_1^p, u_2^p)
2. fix u_2 and optimize by u_1
3. in parallel fix u_1 and optimize for u_2
4. take the convex combination of the results with the weights w_1, w_2

5. repeat

Step 2 and 3 can be further expressed by taking the partial derivatives as

$$\frac{\partial V(u_1^{p+1}, u_2^p)}{\partial u_1} = 0 \tag{3.19}$$

and

$$\frac{\partial V(u_1^p, u_2^{p+1})}{\partial u_2} = 0 \tag{3.20}$$

these expressions then result in

$$\frac{\partial V(u_1^{p\star}, u_2^p)}{\partial u_1} = \begin{pmatrix} u_1^{p\star\top} & u_2^{p\top} \end{pmatrix} \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} c_1^\top & c_2^\top \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{3.21}$$

$$= u_1^{p\star\top} H_{11} + u_2^{p\top} H_{12}^\top + c_1^\top \tag{3.22}$$

$$= 0 \tag{3.23}$$

therefore

$$u_1^{p\star} = -H_{11}^{-1}(H_{12} u_2^p + c_1) \tag{3.24}$$

The same process is done to find the optimal u_2 fixing u_1

$$\frac{\partial V(u_1^p, u_2^{p+1})}{\partial u_1} = \begin{pmatrix} u_1^{p\top} & u_2^{p\star\top} \end{pmatrix} \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} c_1^\top & c_2^\top \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{3.25}$$

$$= u_1^{p\top} H_{12} + u_2^{p\star\top} H_{22} + c_2^\top \tag{3.26}$$

$$= 0 \tag{3.27}$$

therefore

$$u_2^{p\star} = -H_{22}^{-1}(H_{12}^\top u_1^p + c_2) \tag{3.28}$$

Then combining these with weights (w_1, w_2) gives

$$\begin{pmatrix} u_1^{p+1} \\ u_2^{p+1} \end{pmatrix} = w_1 \begin{pmatrix} u_1^{p\star}(u_2^p) \\ u_2^p \end{pmatrix} + w_2 \begin{pmatrix} u_1^p \\ u_2^{p\star}(u_1^p) \end{pmatrix} \tag{3.29}$$

which can be expressed as

$$\begin{pmatrix} u_1^{p+1} \\ u_2^{p+1} \end{pmatrix} = w_1 \begin{pmatrix} -H_{11}^{-1} H_{12} \\ I \end{pmatrix} u_2^p + w_2 \begin{pmatrix} I \\ -H_{22}^{-1} H_{12}^\top \end{pmatrix} u_1^p + \begin{pmatrix} -w_1 H_{11}^{-1} c_1 \\ -w_2 H_{22}^{-1} c_2 \end{pmatrix} \tag{3.30}$$

Which can be rewritten as

$$\begin{pmatrix} u_1^{p+1} \\ u_2^{p+1} \end{pmatrix} = \begin{pmatrix} w_2 I & -w_1 H_{11}^{-1} H_{12} \\ -w_2 H_{22}^{-1} H_{12}^\top & w_1 I \end{pmatrix} \begin{pmatrix} u_1^p \\ u_2^p \end{pmatrix} + \begin{pmatrix} -w_1 H_{11}^{-1} c_1 \\ -w_2 H_{22}^{-1} c_2 \end{pmatrix} \tag{3.31}$$

which has the form

$$u^{p+1} = A u^p + b \tag{3.32}$$

with

$$A = \begin{pmatrix} w_2 I & -w_1 H_{11}^{-1} H_{12} \\ -w_2 H_{22}^{-1} H_{12}^\top & w_1 I \end{pmatrix} \quad b = \begin{pmatrix} -w_1 H_{11}^{-1} c_1 \\ -w_2 H_{22}^{-1} c_2 \end{pmatrix} \tag{3.33}$$

3.2.1. Convergence Proof

In order to prove convergence of the method the task is to prove that the eigenvalues of A are contained within the unit disk. The main tool that is utilized is the property of the Schur complement on the positive definite property Zhang, 2005. This property is true for a symmetric matrix, therefore, first A is transformed with a similarity transformation as Equation 3.34 and Equation 3.35. Since it is similarity transformation the eigenvalues of the newly obtained matrix A_s are the same as of A.

$$T = \begin{pmatrix} \sqrt{w_2} H_{11}^{-1/2} & 0 \\ 0 & \sqrt{w_1} H_{22}^{-1/2} \end{pmatrix} \tag{3.34}$$

$$A_s = T^{-1} A T = \begin{pmatrix} w_2 I & -\sqrt{w_1 w_2} H_{11}^{-1/2} H_{12} H_{22}^{-1/2} \\ -\sqrt{w_1 w_2} H_{22}^{-1/2} H_{12}^{\top} H_{11}^{-1/2} & w_1 I \end{pmatrix} \tag{3.35}$$

Knill, 2019 states that the eigenvalues of a symmetric matrix are real, meaning that there is no imaginary part in them. Thus there is no attempted proof on the imaginary part of the eigenvalues as they are not present for symmetric, and thus by extension, positive or negative semi definite or definite matrices.

Zhang, 2005 states Equation 3.36 which is satisfied as w_2 is a positive scalar, therefore $A_s \succeq 0$ meaning that Equation 3.37 holds. Since the eigenvalues of A_s and A are the same, it is proven that all eigenvalues of A are larger or equal to 0.

$$A_s \succeq 0 \Leftarrow w_2 I \succ 0 \tag{3.36}$$

$$eig(A_s) \geq 0 \; \forall \lambda \tag{3.37}$$

In order to prove that the eigenvalues of A are less than one the M matrix is constructed as Equation 3.38.

$$M = I - A_s = \begin{pmatrix} (1 - w_2) I & \sqrt{w_1 w_2} H_{11}^{-1/2} H_{12} H_{22}^{-1/2} \\ \sqrt{w_1 w_2} H_{22}^{-1/2} H_{12}^{\top} H_{11}^{-1/2} & (1 - w_1) I \end{pmatrix} \tag{3.38}$$

Zhang, 2005 states that M is positive definite given that the first block diagonal element is positive definite and the Schur complement is positive definite. The first condition is met since $w_2 < 1$, therefore $1 - w_2 > 0$. For the second condition the Schur complement is expressed as Equation 3.40.

$$M \succ 0 \Leftrightarrow M_{11} \succ 0, M/M_{11} = S \succ 0 \tag{3.39}$$

$$S = w_2 I - w_2 \, H_{22}^{-1/2} H_{12}^{\top} H_{11}^{-1} H_{12} H_{22}^{-1/2} \tag{3.40}$$

Therefore to show that $S \succ 0$ is the same as showing that

$$I \succ Q \tag{3.41}$$

with Q being positive semi definite as it is constructed with a positive definite matrix in the middle and is multiplied with the transpose of the right side on the left side.

$$Q = H_{22}^{-1/2} H_{12}^{\top} H_{11}^{-1} H_{12} H_{22}^{-1/2} \succeq 0, \tag{3.42}$$

Since $H \succ 0$, it follows from the Schur complement condition for positive definiteness that

$$H_{12}^{\top} H_{11}^{-1} H_{12} \prec H_{22} \tag{3.43}$$

Multiplying buy $H_{22}^{-1/2}$ from both the left and right side leads to

$$H_{22}^{-1/2} H_{12}^{\top} H_{11}^{-1} H_{12} H_{22}^{-1/2} \prec I, \tag{3.44}$$

Meaning that $I - Q \succ 0$ thus $S \succ 0$ holds. Thus $M \succ 0$ holds. Since $M = I - A_s$. due to A_s being symmetric, the eigenvalues of M can be expressed as Horn and Johnson, 1985

$$eig(M) = eig(I) - eig(A_s) \tag{3.45}$$

and also since M is positive definite the eigenvalues are all positive

$$eig(M) > 0 \tag{3.46}$$

Thus

$$1 - eig(A_s) > 0 \tag{3.47}$$

meaning

$$eig(A_s) < 1 \tag{3.48}$$

3.2.2. Equivalence to centralized optimization

Given the problem

$$\frac{1}{2}u^\top Hu + c^\top u + d \tag{3.49}$$

the solution is found by the same method of taking the partial derivative and fining $u^\star$ where it is zero, which produces the solution

$$u^\star = -H^{-1}c \tag{3.50}$$

The convergence of the update Equation 3.32 means that $u_p^\star$ is a fixed point

$$u_p^\star = Au_p^\star + b \tag{3.51}$$

Rearranged for $u^\star$ gives

$$u_p^\star = (I - A)^{-1}b \tag{3.52}$$

therefore the statement says that

$$(I - A)^{-1}b = -H^{-1}c \tag{3.53}$$

expanding on b gives the form

$$(I - A)^{-1}b = (I - A)^{-1}\begin{pmatrix} -w_1H_{11}^{-1} & 0 \\ 0 & -w_2H_{22}^{-1} \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \tag{3.54}$$

which then results in

$$(I - A)^{-1}\begin{pmatrix} -w_1H_{11}^{-1} & 0 \\ 0 & -w_2H_{22}^{-1} \end{pmatrix}c = -H^{-1}c \tag{3.55}$$

which is true for all c if and only if

$$(I - A)^{-1}\begin{pmatrix} -w_1H_{11}^{-1} & 0 \\ 0 & -w_2H_{22}^{-1} \end{pmatrix} = -H^{-1} \tag{3.56}$$

which can be rewritten by taking the inverse as

$$\begin{pmatrix} -1/w_1H_{11} & 0 \\ 0 & -1/w_2H_{22} \end{pmatrix}(I - A) = -H \tag{3.57}$$

expanding on A and H and multiplying both sides by minus 1 provides the relation

$$\begin{pmatrix} 1/w_1H_{11} & 0 \\ 0 & 1/w_2H_{22} \end{pmatrix}\begin{pmatrix} (1 - w_2)I & w_1H_{11}^{-1}H_{12} \\ w_2H_{22}^{-1}H_{12}^\top & (1 - w_1)I \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} \tag{3.58}$$

Computing the matrix product on the left hand side results in

$$\begin{pmatrix} \frac{1-w_2}{w_1}H_11 & H_{11}^{-1}H_{11}H_{12} \\ H_{22}H_{22}^{-1}H_{12}^\top & \frac{1-w_1}{w_2}H_{22} \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} \tag{3.59}$$

The weights are constructed such that the combination is a convex combination, meaning $w_1 + w_2 = 1$, therefore $w_2 = 1 - w_1$. In addition to this the simplification of the inverses multiplied by the matrix result in

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} \tag{3.60}$$

Assuming the statement is true resulted in an identity, therefore the statement is true.

3.3. Problem 3 - Convergence rate

Given the parallel updating of $V(u)$ converges, the convergence rate is expressed in this section. In order to simplify the model the coordinate system is shifted to $z := u - u^\star$ such that the new system has the optimum at it's origin. This simplifies the update of the optimizer as Equation 3.61 with the expression of the fixed point $u^\star = Au^\star + b$, meaning $b = (I - A)u^\star$. The result is in line with the intuition that the solution asymptotically converges to the optimum.

$$u^{p+1} = Au^p + b \tag{3.61}$$
$$z^{p+1} + u^\star = A(z^p + u^\star) + b \tag{3.62}$$
$$z^{p+1} + u^\star = Az^p + Au^\star - Au^\star + u^\star \tag{3.63}$$
$$z^{p+1} = Az^p \tag{3.64}$$

Therefore the difference in the evolution of the cost function is given as Equation 3.65 where $c^\top$ is expressed by the optimality $c^\top = -Hu^\star$

$$V(u^{p+1}) - V(u^p) = \tag{3.65}$$
$$V(z^{p+1} + u^\star) - V(z^p + u^\star) = \tag{3.66}$$
$$V(Az^p + u^\star) - V(z^p + u^\star) = \tag{3.67}$$
$$\frac{1}{2}(Az^p + u^\star)^\top H(Az^p + u^\star) + c^\top(Az^p + u^\star) - \frac{1}{2}(z^p + u^\star)^\top H(z^p + u^\star) - c^\top(z^p + u^\star) = \tag{3.68}$$
$$\frac{1}{2}z^{p\top}(A^\top HA - H)z^p + u^{\star\top}H(A - I)z^p + c^\top(A - I)z^p = \tag{3.69}$$
$$\frac{1}{2}z^{p\top}(A^\top HA - H)z^p + u^{\star\top}H(A - I) - Hu^\star(A - I)z^p = \tag{3.70}$$
$$\frac{1}{2}z^{p\top}(A^\top HA - H)z^p = \tag{3.71}$$
$$\frac{1}{2}(u^p - u^\star)^\top(A^\top HA - H)(u^p - u^\star) = \tag{3.72}$$
$$-\frac{1}{2}(u^p - u^\star)^\top P(u^p - u^\star) \tag{3.73}$$

If $P := -A^\top HA + H$ is positive definite then the reduction of the cost function in each step that is not in the optimum is guaranteed. Given $H \succ 0$ and A is non singular by definition the condition can be met. The following LMI is worked out in detail in the following steps

$$H - A^\top HA \succ 0 \tag{3.74}$$

At this stage H an A are partitioned such that the LMI is expanded to

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} - \begin{pmatrix} w_2 I & -w_1 H_{11}^{-1} H_{12} \\ -w_2 H_{22}^{-1} H_{12}^\top & w_1 I \end{pmatrix}^\top \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} \begin{pmatrix} w_2 I & -w_1 H_{11}^{-1} H_{12} \\ -w_2 H_{22}^{-1} H_{12}^\top & w_1 I \end{pmatrix} \succ 0 \tag{3.75}$$

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} - \begin{pmatrix} w_2 I & -w_2 H_{22}^{-1} H_{12} \\ -w_1 H_{11}^{-1} H_{12}^\top & w_1 I \end{pmatrix} \begin{pmatrix} w_2(H_{11} - H_{12} H_{22}^{-1} H_{12}^\top) & 0 \\ 0 & w_1(H_{22} - H_{12}^\top H_{11}^{-1} H_{12}) \end{pmatrix} \succ 0 \tag{3.76}$$

$$\begin{pmatrix} H_{11} - w_2^2(H_{11} - H_{12} H_{22}^{-1} H_{12}^\top) & H_{12} + w_2 w_1 H_{22}^{-1} H_{12}(H_{22} - H_{12}^\top H_{11}^{-1} H_{12}) \\ H_{12}^\top + w_1 w_2 H_{11}^{-1} H_{12}^\top (H_{11} - H_{12} H_{22}^{-1} H_{12}^\top) & H_{22} - w_1^2(H_{22} - H_{12}^\top H_{11}^{-1} H_{12}) \end{pmatrix} \succ 0 \tag{3.77}$$

According to the assignment it is tasked to show equivalence between Equation 3.77 and Equation 3.78, Equation 3.78 is also expanded in Equation 3.83. Equation 3.88 shows that the two matrices are the same.

$$P = HD^{-1}\tilde{H}D^{-1}H, \quad \tilde{H} = D - N, \quad H = D + N \tag{3.78}$$
$$D^{-1} = \begin{pmatrix} w_1 H_{11}^{-1} & 0 \\ 0 & w_2 H_{22}^{-1} \end{pmatrix}, \quad N = \begin{pmatrix} -w_1^{-1} w_2 H_{11} & H_{12} \\ H_{21} & -w_1 w_2^{-1} H_{22} \end{pmatrix} \tag{3.79}$$

$$H = \begin{pmatrix} H_{11} & H_{12} \\ H_{12}^\top & H_{22} \end{pmatrix} \tag{3.80}$$

$$\tilde{H} = \begin{pmatrix} w_1^{-1}(1+w_2)H_{11} & -H_{12} \\ -H_{12}^\top & w_2^{-1}(1+w_1)H_{22} \end{pmatrix} \tag{3.81}$$

$$\tag{3.82}$$

$$P = HD^{-1}\tilde{H}D^{-1}H \tag{3.83}$$

$$P = HD^{-1}\tilde{H}\begin{pmatrix} w_2 & w_2^{-1}H_{12}H_{22} \\ w_1^{-1}H_{12}^\top H_{11} & w_2^{-1}H_{22}^2 \end{pmatrix} \tag{3.84}$$

$$P = HD^{-1}\begin{pmatrix} (1+w_2)H_{11} - w_2H_{12}H_{22}^{-1}H_{12}^\top & H_{12} \\ H_{12}^\top & -w_1H_{12}^\top H_{11}^{-1}H_{12} + (1+w_1)H_{22} \end{pmatrix} \tag{3.85}$$

$$P = H\begin{pmatrix} w_1(1+w_2)I - w_1w_2H_{11}^{-1}H_{12}H_{22}^{-1}H_{12}^\top & w_1H_{11}^{-1}H_{12} \\ w_2H_{22}^{-1}H_{12}^\top & -w_1w_2H_{22}^{-1}H_{12}^\top H_{11}^{-1}H_{12} + w_2(1+w_1)I \end{pmatrix} \tag{3.86}$$

$$P = \begin{pmatrix} w_1(1+w_2)H_{11} + w_2(1-w_1)H_{12}H_{22}^{-1}H_{12}^\top & \big(w_1 + w_2(1+w_1)\big)H_{12} - w_1w_2H_{12}H_{22}^{-1}H_{12}^\top H_{11}^{-1}H_{12} \\ \big(w_1(1+w_2)+w_2\big)H_{12}^\top - w_1w_2H_{12}^\top H_{11}^{-1}H_{12}H_{22}^{-1}H_{12}^\top & w_1(1-w_2)H_{12}^\top H_{11}^{-1}H_{12} + w_2(1+w_1)H_{22} \end{pmatrix} \tag{3.87}$$

$$P = \begin{pmatrix} (1-w_2^2)H_{11} + w_2^2H_{12}H_{22}^{-1}H_{12}^\top & (w_1+1)H_{12} - w_1w_2H_{12}H_{22}^{-1}H_{12}^\top H_{11}^{-1}H_{12} \\ (w_2+1)H_{12}^\top - w_1w_2H_{12}^\top H_{11}^{-1}H_{12}H_{22}^{-1}H_{12}^\top & w_1^2H_{12}^\top H_{11}^{-1}H_{12} + (1-w_1^2)H_{22} \end{pmatrix} \tag{3.88}$$

Where the simplifications were used:$w_1(1+w_2) = (1-w_2)(1+w_2) = 1-w_2^2$, $w_2(1-w_1) = w_2^2$.

The last task of this part is to prove that P is positive definite. A matrix is positive definite if and only if it can be written in the form

$$P_p = P_1^\top P_1 \tag{3.89}$$

this is due to the definition of the property positive definite

$$x^\top P_p x = x^\top P_1^\top P_1 x = y^\top y \geq 0 \tag{3.90}$$

With the definitions of H and D, both of them are symmetric meaning $H = H^\top$, $D^{-1} = D^{-1\top}$, therefore P can be written in the form

$$P = H^\top D^{-1\top}\tilde{H}D^{-1}H \tag{3.91}$$

where $\tilde{H}$ can be written as

$$\tilde{H} = \begin{pmatrix} H_{11} & -H_{12} \\ -H_{12}^\top & H_{22} \end{pmatrix} + \begin{pmatrix} \frac{2w_2}{1-w_2}H_{11} & 0 \\ 0 & \frac{2w_1}{1-w_1}H_{22} \end{pmatrix} \tag{3.92}$$

Since H is given to be positive definite it's diagonal blocks are also positive definite therefore the second part of the addition is positive definite. As provided in the assignment, given H is positive definite so is Equation 3.93. Since the additional of two positive definite matrices is also a positive definite matrix $\tilde{H}$ is positive definite. Since the rest of P is written in with the transpose multiplied by itself, it is shown that P is positive definite. Thus proves asymptotic reduction of the cost function, and this convergence of iteration $u^{p+1}(u^p)$

$$\bar{H} = \begin{pmatrix} H_{11} & -H_{12} \\ -H_{12}^\top & H_{22} \end{pmatrix} \succ 0 \tag{3.93}$$

3.4. Parallel optimization without convex combination

In this section the problem is reformulated with 3 parallel computations.

$$V(u_1,u_2,u_3) = \frac{1}{2}\begin{pmatrix} u_1^\top & u_2^\top & u_3^\top \end{pmatrix}\begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{12}^\top & H_{22} & H_{23} \\ H_{13}^\top & H_{23}^\top & H_{33} \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + \begin{pmatrix} c_1^\top & c_2^\top & c_3^\top \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + d \tag{3.94}$$

taking the partial derivative with respect to the partitioned vector u gives

$$\frac{\partial V}{\partial u_1} = \begin{pmatrix} H_{11} & H_{12} & H_{13} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + c_1^\top \tag{3.95}$$

$$\frac{\partial V}{\partial u_2} = \begin{pmatrix} H_{12}^\top & H_{22} & H_{23} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + c_2^\top \tag{3.96}$$

$$\frac{\partial V}{\partial u_3} = \begin{pmatrix} H_{13}^\top & H_{23}^\top & H_{33} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} + c_3^\top \tag{3.97}$$

$$\tag{3.98}$$

finding the optimum for each of them by equating the derivative to 0. The conditions for this still hold: The set of u needs to be convex and the function is by definition convex, as it is a quadratic.

$$u_1^{p+1} = H_{11}^{-1} \left(- \begin{pmatrix} H_{12} & H_{13} \end{pmatrix} \begin{pmatrix} u_2^p \\ u_3^p \end{pmatrix} - c_1^\top \right) \tag{3.99}$$

$$u_2^{p+1} = H_{22}^{-1} \left(- \begin{pmatrix} H_{12}^\top & H_{23} \end{pmatrix} \begin{pmatrix} u_1^p \\ u_3^p \end{pmatrix} - c_2^\top \right) \tag{3.100}$$

$$u_3^{p+1} = H_{33}^{-1} \left(- \begin{pmatrix} H_{13}^\top & H_{23}^\top \end{pmatrix} \begin{pmatrix} u_1^p \\ u_2^p \end{pmatrix} - c_3^\top \right) \tag{3.101}$$

There is no convex combination step taken at this approach therefore the update is described as

$$u^{p+1} = \begin{pmatrix} 0 & -H_{11}^{-1} H_{12} & -H_{11}^{-1} H_{13} \\ -H_{22}^{-1} H_{12}^\top & 0 & -H_{22}^{-1} H_{23} \\ -H_{33}^{-1} H_{13}^\top & -H_{33}^{-1} H_{23}^\top & 0 \end{pmatrix} u^p - \begin{pmatrix} H_{11}^{-1} c_1^\top \\ H_{22}^{-1} c_2^\top \\ H_{33}^{-1} c_3^\top \end{pmatrix} \tag{3.102}$$

Applying this to the provided example of Equation 3.103 the resulting update is given as Equation 3.104, which provides an increase in the cost function, from 4 to 6.75, thus a counter example has been found, therefore the provided method's convergence is disproven. Due to this result it becomes clear that the convex combination is a necessary step as it is hypothesized that update including the convex combination would yield positive result.

$$H = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad u^0 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \tag{3.103}$$

$$u^1 = \begin{pmatrix} -0.5 \\ -3 \\ -1 \end{pmatrix} \tag{3.104}$$

4

Distributed Constrained LQR

In this task the distributed discrete LQR of an LTI system is presented. The system is defined with

$$x_i(t+1) = A_i x_i(t) + B_i u_i(t) \tag{4.1}$$

$$\forall i \in \mathbb{I} := \{1:4\} \tag{4.2}$$

$$\forall t \in \mathbb{T} := \{0:T-1\} \tag{4.3}$$

Since the objective is to find the optimal input action the dynamics is rewritten as

$$x_i(t) = A_i^t x_{i,o} + \sum_{j\in\{0:t-1\}} A_i^{t-1-j} B_i u_i(j) \tag{4.4}$$

Note: A_i^T does not mean the transpose, in this report $^\top$ indicates the transpose, but it means A_i on the power T, which is the horizon, the number of steps the optimization optimizes for.

$$x_{i,1:T} = \bar{A}_i x_{i,0} + \bar{B}_i u_{i,0:T-1} \tag{4.5}$$

where

$$x_{i,1:T} = \begin{pmatrix} x_i(1) \\ \vdots \\ x_i(T) \end{pmatrix} \quad u_{i,0:T-1} = \begin{pmatrix} u_i(0) \\ \vdots \\ u_i(T-1) \end{pmatrix} \tag{4.6}$$

and

$$\bar{A}_i = \begin{pmatrix} A_i \\ A_i^2 \\ \vdots \\ A_i^T \end{pmatrix} \quad \bar{B}_i = \begin{pmatrix} B_i & & & \\ A_i B_i & B_i & & \\ \vdots & & \ddots & \\ A_i^{T-1} B_i & \cdots & A_i B_i & B_i \end{pmatrix} \tag{4.7}$$

coupling constraint

$$x_i(T) = x_f \forall i \in \mathbb{I} \tag{4.8}$$

decoupled constraint

$$|u_i(t)| \leq u_{max} \forall i \in \mathbb{I}\ \forall t \in \mathbb{T} \tag{4.9}$$

Loss function, since the first state is given, it is not influenced by the control input, it is not included in the computation.

$$J(u_{1:4,0:T-1}, x(u_{1:4,0:T-1})) = \sum_{i\in\mathbb{I}} \sum_{t\in\mathbb{T}} x_i(t+1)^\top x_i(t+1) + u_i(t)^\top u_i(t) \tag{4.10}$$

$$J(u_{1:4,0:T-1}) = \sum_{i\in\mathbb{I}} J_i(u_{i,0:T-1}) \tag{4.11}$$

The coupling constraint can be expressed as

$$\lambda^{i,i+1}(x_{i,T}(u_{i,0:T-1}) - x_{i+1,T}(u_{i+1,0:T-1})) = 0 \tag{4.12}$$

$$with \qquad x_{i,T}(u_{i,0:T-1}) = A_i^T x_{i,0} + \tilde{B}_i u_{i,0:T-1} \tag{4.13}$$

$$x_{i+1,T}(u_{i+1,0:T-1}) = A_{i+1}^T x_{i+1,0} + \tilde{B}_{i+1} u_{i+1,0:T-1}) \tag{4.14}$$

$$where \qquad \tilde{B}_i = \begin{pmatrix} A_i^{T-1}B_i & \cdots & A_iB_i & B_i \end{pmatrix} \tag{4.15}$$

therefore the Lagrangian of problem is formulated as

$$u^\star_{1:4,0:T-1} = \arg\min_{u_{1:4,0:T-1}} \sum_{i\in\mathbb{I}} \|\bar{A}_i x_{i,0} + \bar{B}_i u_{i,0:T-1}\|_2^2 + \|u_{i,0:T-1}\|_2^2 \tag{4.16}$$

$$+ \sum_{i=1}^{3} \lambda^{i,i+1}(A_i^T x_{i,0} + \tilde{B}_i u_{i,0:T-1} - A_{i+1}^T x_{i+1,0} - \tilde{B}_{i+1} u_{i+1,0:T-1}) \tag{4.17}$$

$$s.t. \tag{4.18}$$

$$|u_i(t)| \le u_{max} \forall i \in \mathbb{I}\ \forall t \in \mathbb{T} \tag{4.19}$$

The dual function is separated into parts that only depend on one input sequence $u_{i,0:T-1}$ Equation 4.20. This form allows to distribute the computation into four lower level dual functions Equation 4.25

$$d(\lambda) = \inf_{u_{1,0:T-1}} \left(J_1(u_{1,0:T-1}) + \lambda^{1,2} x_{1,T}(u_{1,0:T-1})\right) + \tag{4.20}$$

$$\inf_{u_{2,0:T-1}} \left(J_2(u_{2,0:T-1}) + \lambda^{2,3} x_{2,T}(u_{2,0:T-1}) - \lambda^{1,2} x_{2,T}(u_{2,0:T-1})\right) + \tag{4.21}$$

$$\inf_{u_{3,0:T-1}} \left(J_3(u_{3,0:T-1}) + \lambda^{3,4} x_{3,T}(u_{3,0:T-1}) - \lambda^{2,3} x_{3,T}(u_{3,0:T-1})\right) + \tag{4.22}$$

$$\inf_{u_{4,0:T-1}} \left(J_4(u_{4,0:T-1}) - \lambda^{3,4} x_{4,T}(u_{4,0:T-1})\right) \tag{4.23}$$

$$s.t. \qquad |u_i(t)| \le u_{max} \forall i \in \mathbb{I}\ \forall t \in \mathbb{T} \tag{4.24}$$

$$d_1(\lambda^{1,2}) = \inf_{u_{1,0:T-1}} \left(J_1(u_{1,0:T-1}) + \lambda^{1,2} x_{1,T}(u_{1,0:T-1})\right) \tag{4.25}$$

$$d_2(\lambda^{2,3}, \lambda^{1,2}) = \inf_{u_{2,0:T-1}} \left(J_2(u_{2,0:T-1}) + \lambda^{2,3} x_{2,T}(u_{2,0:T-1}) - \lambda^{1,2} x_{2,T}(u_{2,0:T-1})\right) \tag{4.26}$$

$$d_3(\lambda^{3,4}, \lambda^{2,3}) = \inf_{u_{3,0:T-1}} \left(J_3(u_{3,0:T-1}) + \lambda^{3,4} x_{3,T}(u_{3,0:T-1}) - \lambda^{2,3} x_{3,T}(u_{3,0:T-1})\right) \tag{4.27}$$

$$d_4(\lambda^{3,4}) = \inf_{u_{4,0:T-1}} \left(J_1(u_{4,0:T-1}) - \lambda^{3,4} x_{4,T}(u_{4,0:T-1})\right) \tag{4.28}$$

$$s.t. \qquad |u_i(t)| \le u_{max} \forall i \in \mathbb{I}\ \forall t \in \mathbb{T} \tag{4.29}$$

Each computing node (i) optimizes by finding the minimum of the separated loss function and translates it into the dual function $d_i(\lambda)$. To enable the optimization the nodes can either return a function of λ, which would explicitly allow the merging of the results, or, it can return the optimum for a given λ which then can be updated, resulting in a process of iteration until convergence. Either way, due to the characteristics of a coupled problem, the results gained by distributing the computation have to go under a merge that enacts the coupling of the coupling constraint. This introduces the next step, solving the textitmaster dual problem Equation 4.30 Keviczky, 2025b

$$\lambda_{k+1} = \arg\max_{\lambda} \sum_{i\in\mathbb{I}} d_i(\lambda(u^\star_{1:4,0:T-1}(\lambda_k))) \tag{4.30}$$

where the quadratic loss part J_i of the dual function doesn't depend on λ therefore it can be written as

$$\lambda_{k+1} = \arg\max_{\lambda} \begin{pmatrix} \lambda^{1,2\top} & \lambda^{2,3\top} & \lambda^{3,4\top} \end{pmatrix} \begin{pmatrix} x_{1,T}(u^\star_{1,0:T-1}(\lambda_k)) - x_{2,T}(u^\star_{2,0:T-1}(\lambda_k)) \\ x_{2,T}(u^\star_{2,0:T-1}(\lambda_k)) - x_{3,T}(u^\star_{3,0:T-1}(\lambda_k)) \\ x_{3,T}(u^\star_{3,0:T-1}(\lambda_k)) - x_{4,T}(u^\star_{4,0:T-1}(\lambda_k)) \end{pmatrix} \tag{4.31}$$

$$=: \arg\max_{\lambda} \lambda^\top v(u^\star_{1:4,0:T-1}(\lambda_k)) \tag{4.32}$$

Since this optimization problem includes information that is computed with respect to λ_k, opposing the explicit formulation, can't be solved in one go. The optimal input values need to be updated which is done in iterations, therefore λ is updated as a steepest descent step with step size α Keviczky, 2025b.

$$\lambda_{k+1} = \lambda_k + \alpha v(u^{\star}_{1:4,0:T-1}(\lambda_k)) \tag{4.33}$$

Given the dynamics of the plants are the same and defined as Equation 4, and they differ at their initial states Equation 4, the process results in the process where the evolution of the terminal state is plotted at Figure 4.1. The process is validated by comparing the result to the centralized optimization. Both methods produce the same result of Equation 4.34

$$A_i = \begin{bmatrix} 0.626 & -0.6975 \\ 0.8719 & 0.626 \end{bmatrix}, \quad B_i = \begin{bmatrix} -0.0748 \\ 0.1744 \end{bmatrix} \quad \forall i \in \mathbb{I}$$

$$x_{0,1} = \begin{bmatrix} -10 \\ -0.5 \end{bmatrix}, \quad x_{0,2} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad x_{0,3} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad x_{0,4} = \begin{bmatrix} 5 \\ -0.5 \end{bmatrix}$$

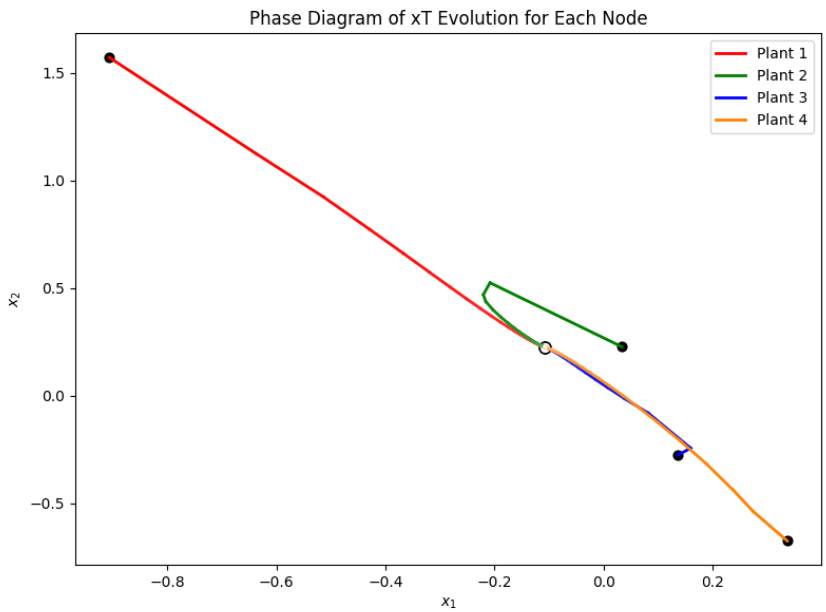

Figure 4.1: Base Case Distributed LQR with $\alpha = 5$

$$x_{i,T} = \begin{pmatrix} -0.1078 \\ 0.2258 \end{pmatrix} \forall i \in \mathbb{I} \tag{4.34}$$

This solution results in the evolution of the states Figure 4.3, which is compared to centralized solution is almost identical with an mean square difference of only $3.3e-11$

In order to show that the distributed process converges the error is plotted with respect to the iterations of the agents. The error is defined as the square of the two norm of the term v defined in Equation 4.32. Given if this goes to zero than the duality gap goes to zero, which proves strong duality and thus validates the process. The error is visualized in a log plot with base 10, Figure 4.4 shows asymptotic convergence, which corresponds to the slides Keviczky, 2025b.

$$e = \|v\|_2^2 \tag{4.35}$$

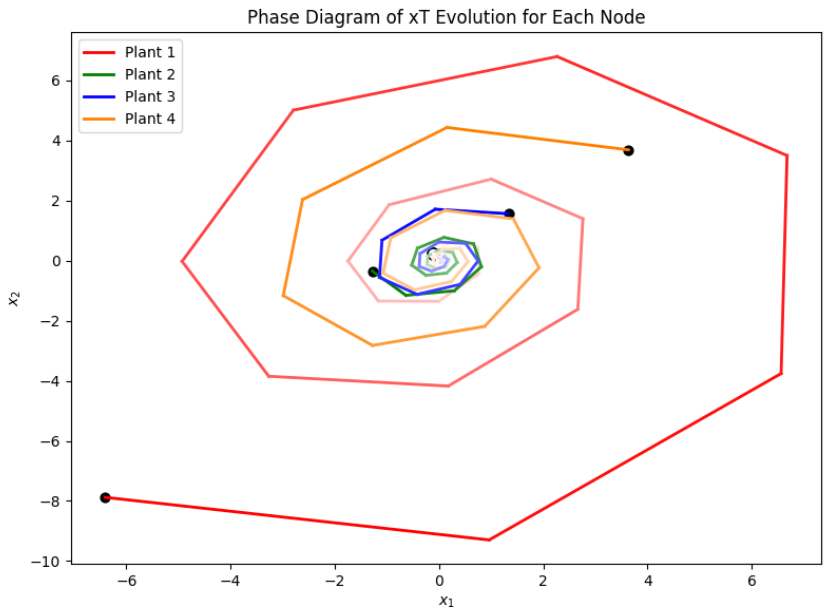

Figure 4.2: Base Case Distributed LQR evolution with $\alpha = 5$

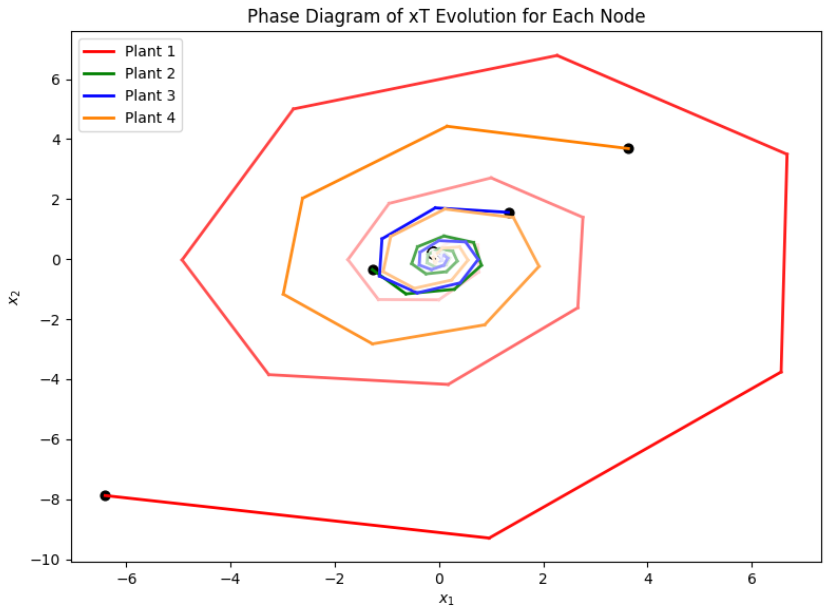

Figure 4.3: Centralized LQR evolution

4.0.1. Effect of step size

In this section the affect of step size is investigated on the evolution of the error over the iterations. First, the effect of changing the static value of a constant α on the range is visualized on Figure 4.5. It shows that with the current grid of α values, the best performing one is with $\alpha = 7$.

It is common practice to implement a variable step size for a steepest descend algorithm. The distributed LQR present in this report is a steepest descend algorithm, therefore the variable step size is implemented for testing purposes. The motivation is that the further the process is developed the optimal update difference size decreases. This is already present for the static gain, as it is a necessary condition for convergence, but that doesn't mean that the process can't be improved by introducing a variable α. $\alpha(k)$ is commonly defined as an exponential decay $\alpha(k) = \alpha_0 r^k$, thus it is described by two variables α_0 and r. Figure 4.5 shows that the fastest converging α is 7, therefore α_0 is fixed to be $\alpha_0 = 7$. The decay rate is tested on the grid $[0.95, 1]$ with 10 different sizes resulting the convergence characteristics Figure 4.6. The plot indicates that the exponential decay decreases the performance,

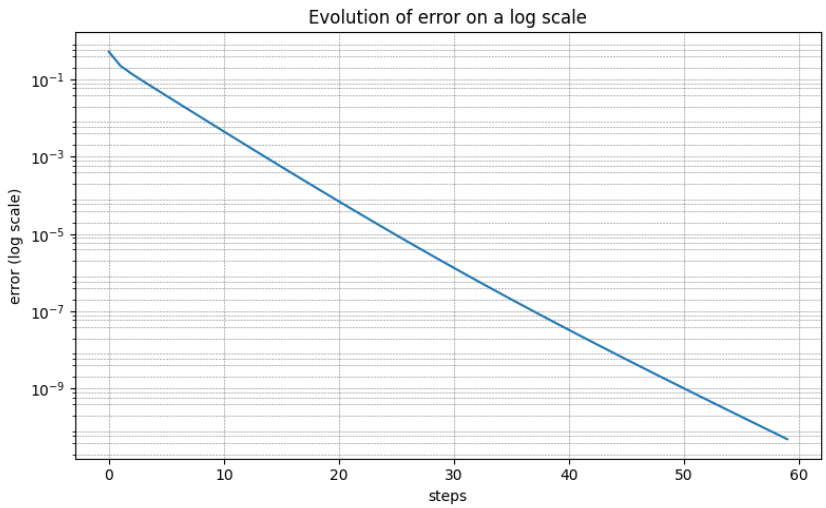

Figure 4.4: Error with respect to the iteration steps on a log scale

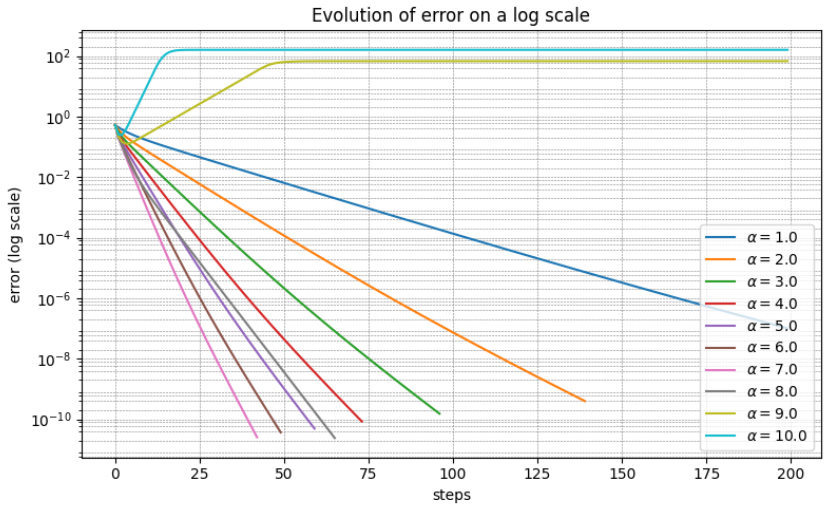

Figure 4.5: Error with respect to the iteration steps on a log scale for static α values

thus it indicates that the ideal α is static.

4.1. Nesterov method

The Nesterov method is implemented according to Nesterov, 1983, which introduces a momentum update to the lambda update Equation 4.36. The motivation is that it can reduce oscillation around the optimum. It produces improvements over $\beta \in [0.15, 0.45]$ Figure 4.7. To further investigate the behaviour, this segment is plotted on Figure 4.8. The figure shows that for a maximum value of $\beta = 0.28$ the convergence happens at an exponential rate, while this breaks for $\beta \geq 0.32$, therefore the optimal value for β is chosen to be 0.28. Thus the convergence increased in rate from reaching the stopping condition from 42 steps to 23 steps, which is a $\approx 45\%$ increase in performance.

$$\lambda^{k+1} = y^k + \alpha v \tag{4.36}$$

$$y^{k+1} = \lambda^{k+1} + \beta\left(\lambda^{k+1} - \lambda^k\right) \tag{4.37}$$

4.2. Consensus based distributed LQR

In this section the distributed problem is reformulated. Previously there were 4 nodes that were doing the separated optimizations while λ was updated via a coordinator node. This approach introduces a

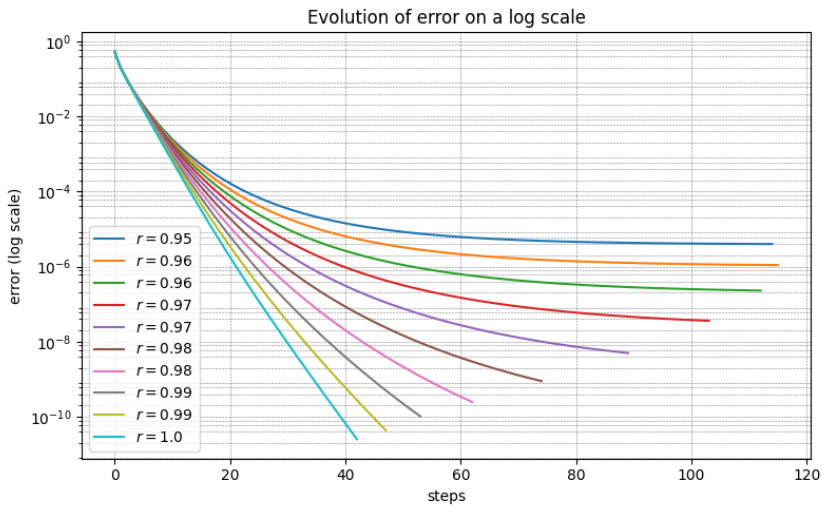

Figure 4.6: Error with respect to the iteration steps on a log scale for dynamic α values

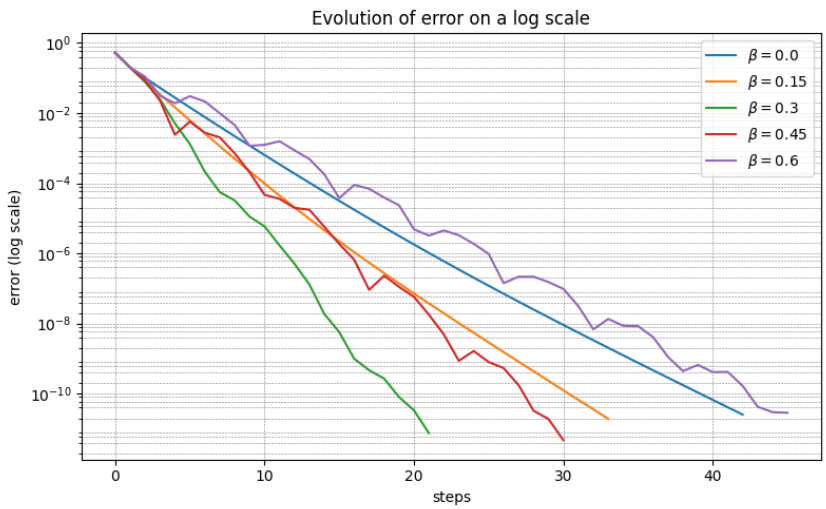

Figure 4.7: Log Error with respect to updates with Nesterov method

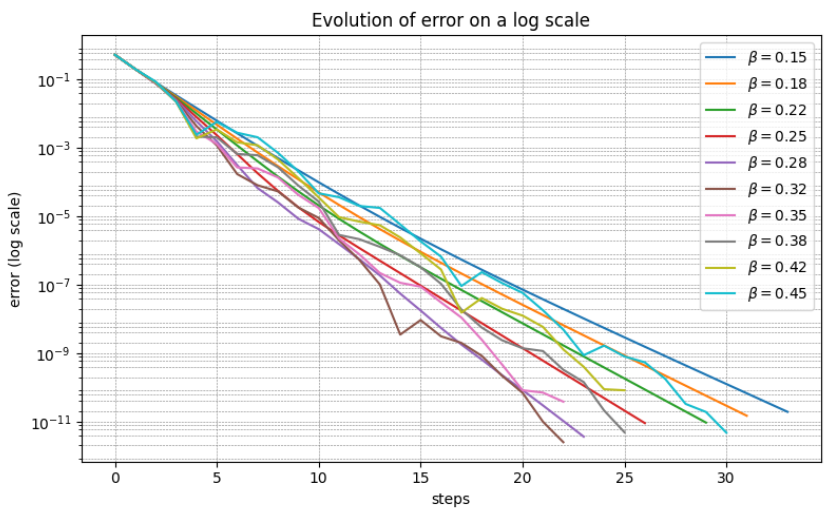

Figure 4.8: Log Error with respect to updates with Nesterov method, second set of β

new node and poses requirements on the network communication. In the following approach there is no centralized coordinator, but the information if exchanged and updated in a distributed manner.

Instead confusing the dual decomposition in this problem the coupling is represented by a coupling variable x_f. x_f is initialized at an arbitrary point, chosen to be the origin in this case, and is updated locally. Each local update produces a local update of x_f, which is denoted as x_f^i. For the update it is necessary to gain the gradient of the optimal loss with respect to the x_f, which is done by using the central difference numerical differentiation Equation 4.38.

$$\nabla J_i^\star(x_f) = \begin{pmatrix} \dfrac{J_i^\star(x_f + \begin{pmatrix} h & 0 \end{pmatrix}^\top) - J_i^\star(x_f - \begin{pmatrix} h & 0 \end{pmatrix}^\top))}{2h} \\ \dfrac{J_i^\star(x_f + \begin{pmatrix} 0 & h \end{pmatrix}^\top)) - J_i^\star(x_f - \begin{pmatrix} 0 & h \end{pmatrix}^\top))}{2h} \end{pmatrix} \tag{4.38}$$

The gradient is used to create the local updates of x_f with the standard gradient descend formula, where α is again the step size. Alternatively the gradient of the optimal cost can be expressed as

$$\nabla J_i^\star = \frac{\partial J_i^\star}{\partial u_{i,0:T-1}^\star} \frac{\partial u_{i,0:T-1}^\star}{\partial x_f} \tag{4.39}$$

$$= 2(\bar{B}^\top(\bar{A}x_0 + \bar{B}u_{i,0:T-1}^\star) + u_{i,0:T-}^\star)(\tilde{B}^\top\tilde{B})^{-1}\tilde{B}^\top \tag{4.40}$$

where $u_{i,0:T-1}^\star$ is computed as the optimal input sequence as defined before at $x_{f,i}^{(k)}$. This alternative offers the reduction of optimizations to be solved from 4 to 1.

$$x_f^i(k+1) = x_f^i(k) - \alpha \nabla J_i(x_f) \tag{4.41}$$

Following the local update took place for every node the results are coupled by the consensus update. In the consensus update the x_f values are updated with respect to a symmetric doubly stochastic matrix W Keviczky, 2025c. The consensus update can is repeated a parameter *c* amount of times.

$$x_f^i = \sum_{j=1}^{4} W(i,j) x_f^j \quad \forall i \in \mathbb{I} \tag{4.42}$$

$$with \qquad W = \begin{pmatrix} 0.75 & 0.25 & 0 & 0 \\ 0.25 & 0.5 & 0.25 & 0 \\ 0 & 0.25 & 0.5 & 0.25 \\ 0 & 0 & 0.25 & 0.75 \end{pmatrix} \tag{4.43}$$

For the different number of consensus updates, $h = 0.1$ and $\alpha = 0.02$ the resulting convergence rates are displayed on a log plot on Figure 4.9. It figure shows that at increasing c improves the convergence, and is even necessary to have at least 6; however, at $c = 33$ the positive influence dissipates. The plot shows an upper bound of of the convergence rate that is very similar to the previously obtained convergence rate of the Nesterov method.

4.3. Alternating Direction Method of Multipliers

In this section a new approach is taken to distribute the optimization problem. Th ADMM approach is implemented according to Keviczky, 2025a. The ADMM consists of 3 computations as Equation 4.44.

$$x_{f,i}^{(k+1)} := \arg\min_{x_{fi}} \left(J_i(x_{fi}) + \lambda_i^{k^\top}(x_{fi} - z^k) + \frac{\rho}{2}\|x_{fi} - z^k\|_2^2 \right) \tag{4.44}$$

$$z^{k+1} := \frac{1}{N}\sum_{i=1}^{N}\left(x_i^{k+1} + \frac{1}{\rho}\lambda_i^k \right) \tag{4.45}$$

$$\lambda_i^{k+1} := \lambda_i^k + \rho\left(x_i^{k+1} - z^{k+1}\right) \tag{4.46}$$

The first one is implemented with a equating the gradient to zero.

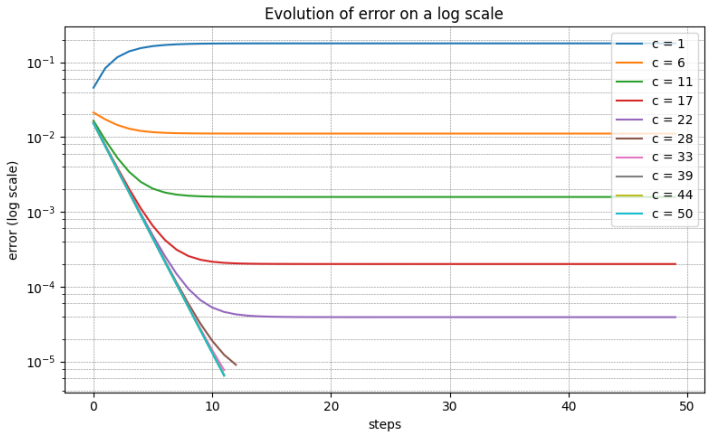

Figure 4.9: Log error of the incremental subgradient method with respect to the number of consensus steps taken = c with $h = 0.1$, $\alpha = 0.02$

$$x_{f,i}^{(k+1)} = z^k - 1/\rho(\nabla J_i(x_{f,i}^{(k)}) + \lambda_i^k) \tag{4.47}$$

The process is repeated 50 times to investigate the behaviour of the estimated terminal states of each node. At $\rho = 100$ and initial guess of $x_f = (1 \quad 1)^\mathsf{T}$ Figure 4.10 shows the evolution of $x_{f,i}$, and Figure 4.11 shows the evolution of the error on a log scale.

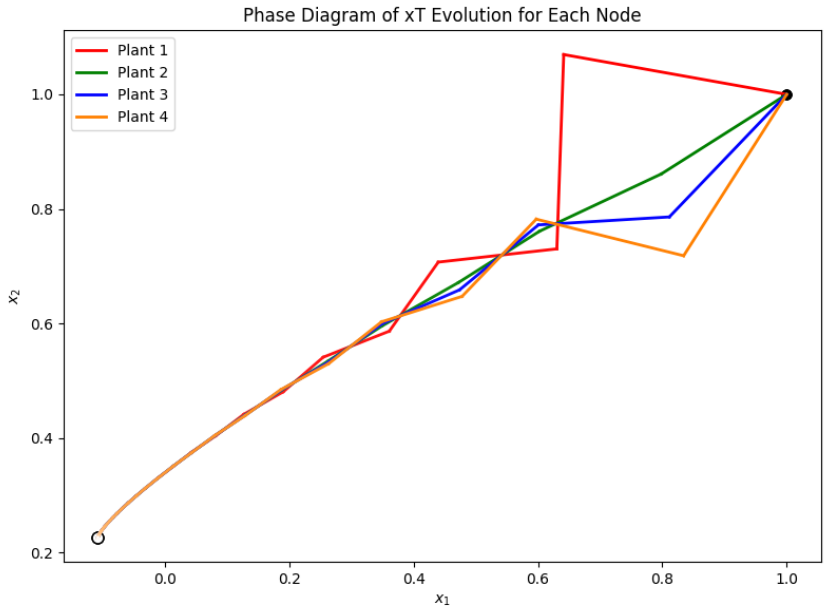

Figure 4.10: Evolution of $x_{f,i}$ during the ADMM process

This process results in the same terminal state Equation 4.48, which then results in the same behaviour of the plants as the distributed controllers before Figure 4.12

$$x_{i,T} = \begin{pmatrix} -0.1078 \\ 0.2258 \end{pmatrix} \forall i \in \mathbb{I} \tag{4.48}$$

4.3.1. Effect of rho

The effect of ρ is investigated by testing the process, resulting in Figure 4.13, which shows that the optimal value is around 60. It also indicates a behaviour that is unique among the methods described in this report, which is that the error is not strictly decreasing over the iterations but increases after attaining the minimum and settles at a suboptimal solution a couple of iterations later.

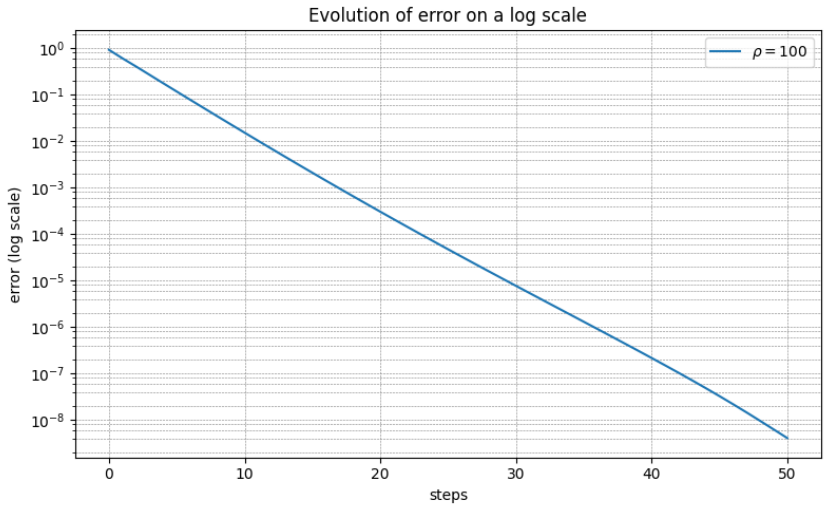

Figure 4.11: Evolution of $x_{f,i}$ during the ADMM process

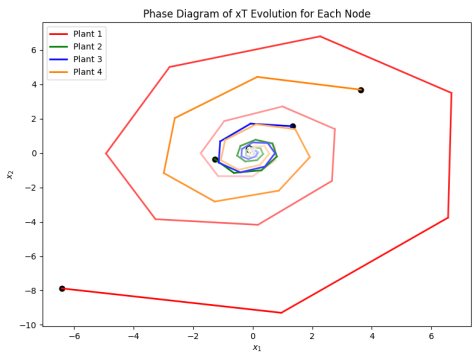

Figure 4.12: Resulting Evolution with ADMM

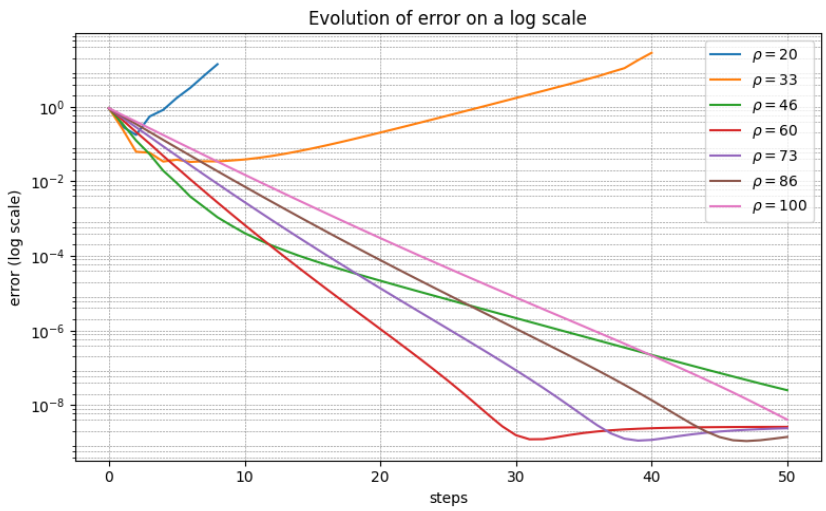

Figure 4.13: ADMM evolution of error with respect to ρ

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